

Computer Vision 1

2-2: Neighborhood Processing, Convolution and Image Filtering

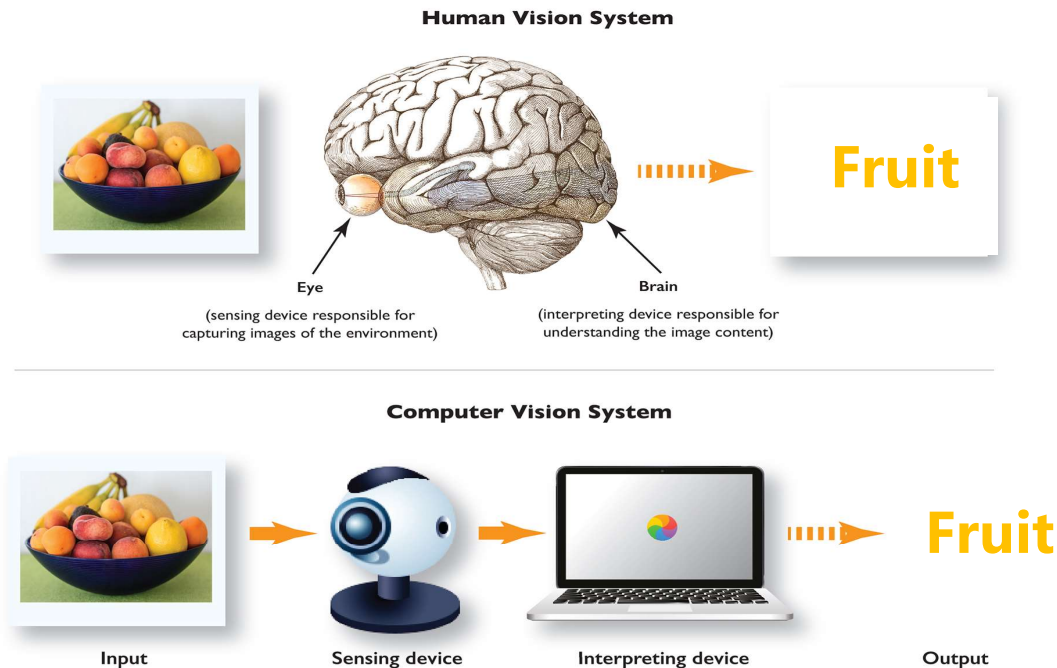
Master of AI – 2024/2025
University of Amsterdam

Arnoud Visser & Martin Oswald

Based on a presentation of Shaodi You,
augmented with material of Linda Shapiro & many others

The computer vision problem

- Make a computer see and understand images



- We know it is physically possible –
we do it every day and effortlessly!

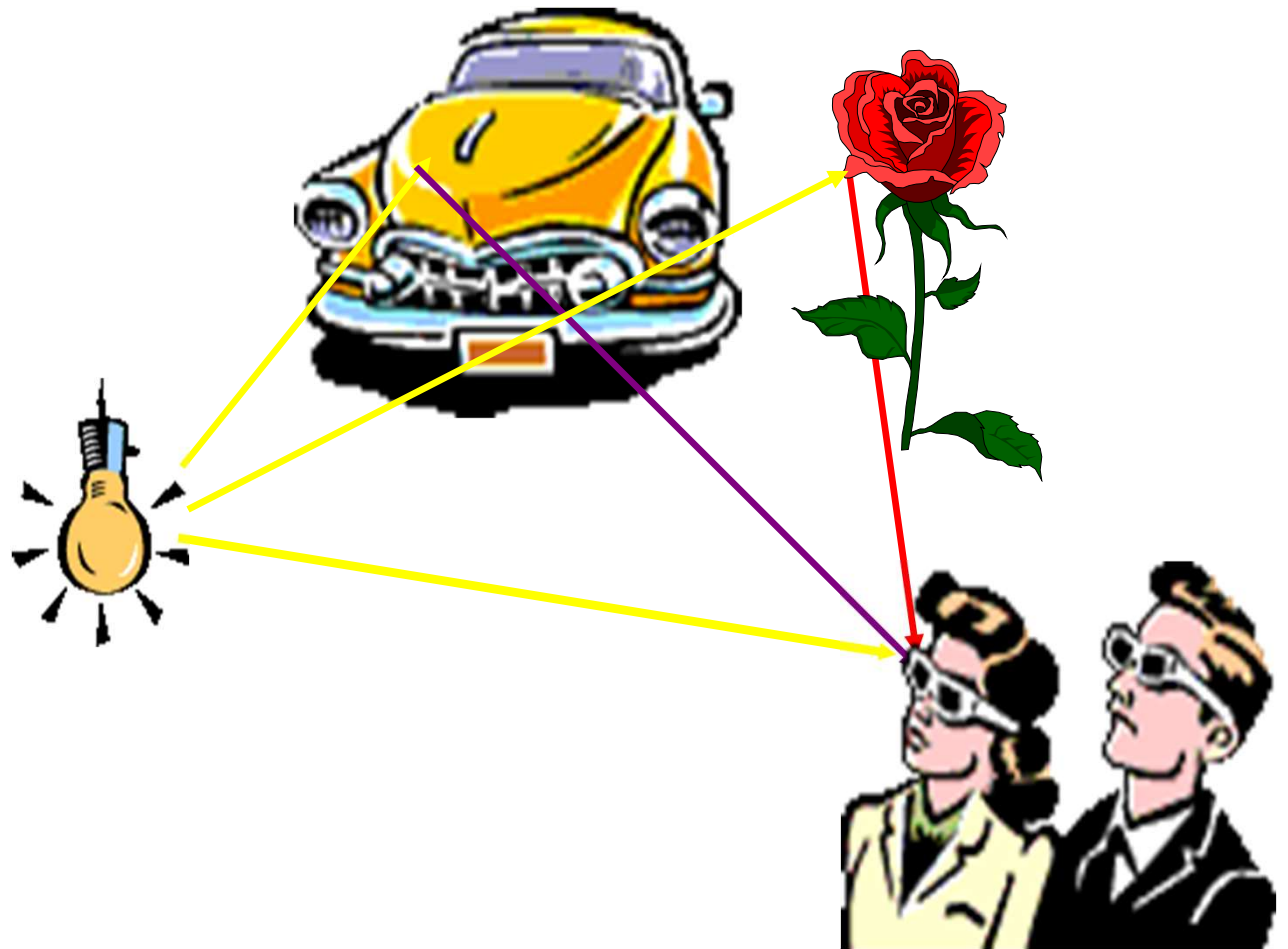
What makes an image?

the triplet light-objects-observer

Light source

Object(s)

Sensor

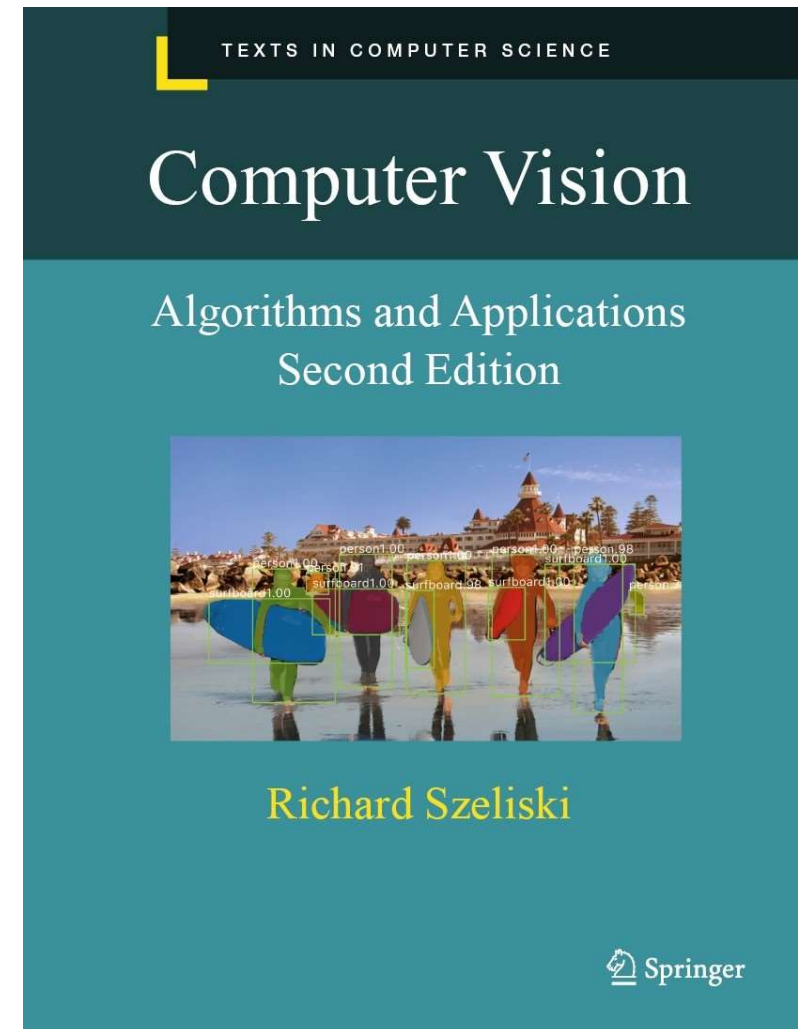


Outline

- Neighborhood image processing
 - Connected Component Analysis (CCA)
 - Mathematical Morphology
 - Dilation
 - Erosion
 - Opening
 - Closing
- Image filtering
 - Convolution
 - Frequently used filters
 - Gaussian filter
 - Nonlinear Filters
 - Median filter
 - Bilateral filter
- Frequency domain analysis

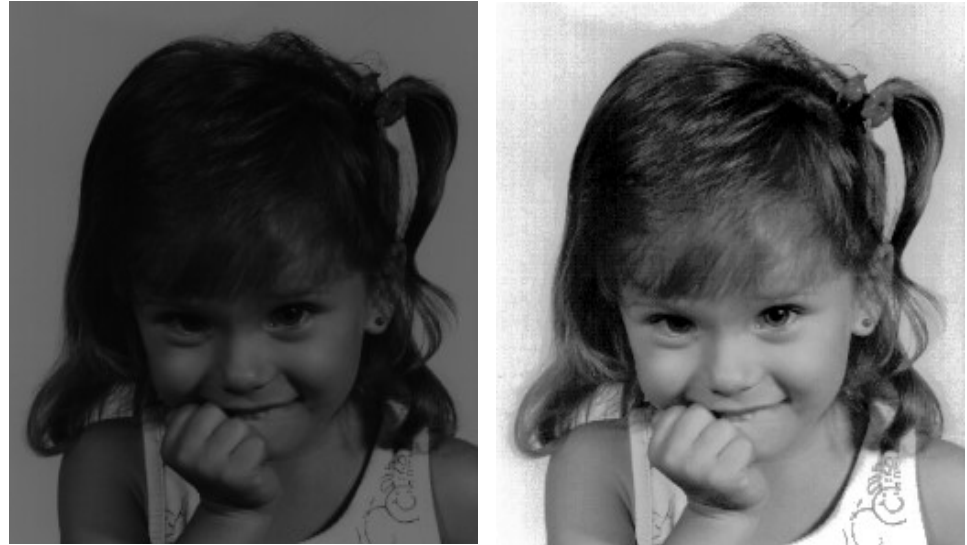
Textbook

- 13.1.1
- 3.2
- 3.3
- 3.4



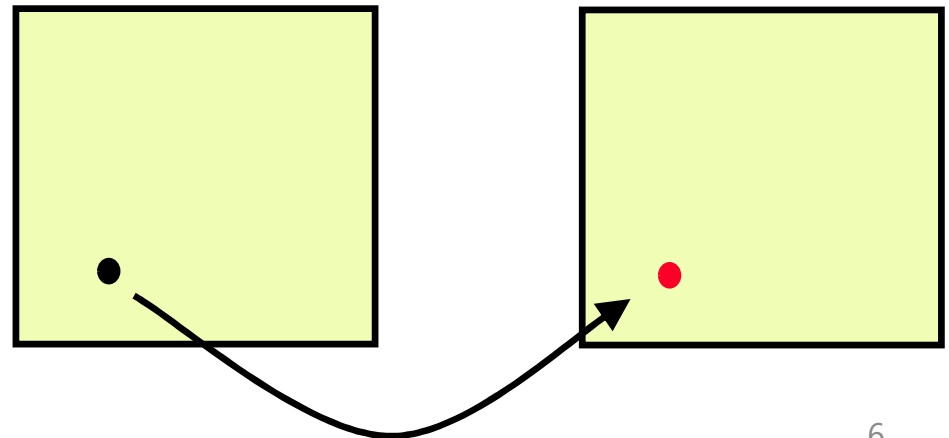
Point Operations

- Operation on ***Pixel's value***
- Context free
- Operation can be performed on the image histogram
- Example:



$$g(x, y) = M(f(x, y))$$

$$g(x, y) = \alpha \cdot f(x, y) + \beta$$



Neighborhood Operations

- Operation depends on ***Pixel's value*** and its spatial ***neighbors***.

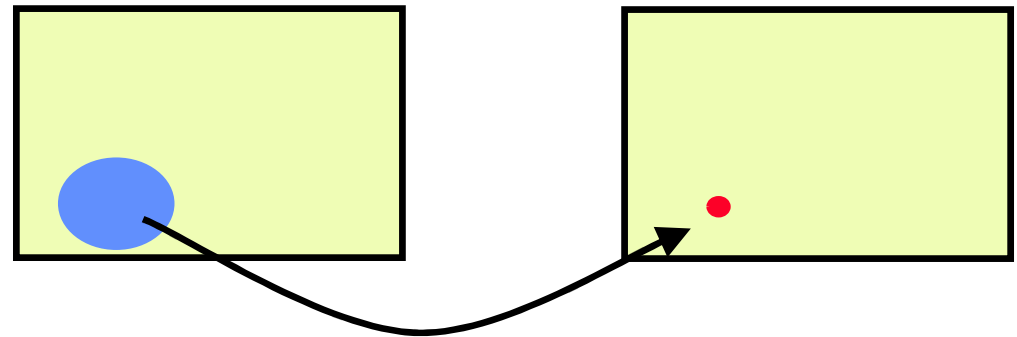
- Context dependent.

- Example:

$$g(x, y) = \sum_{i, j \in N(x, y)} f(i, j) / n$$



$$g(x, y) = M(\{f(i, j) | (i, j) \in N(x, y)\})$$



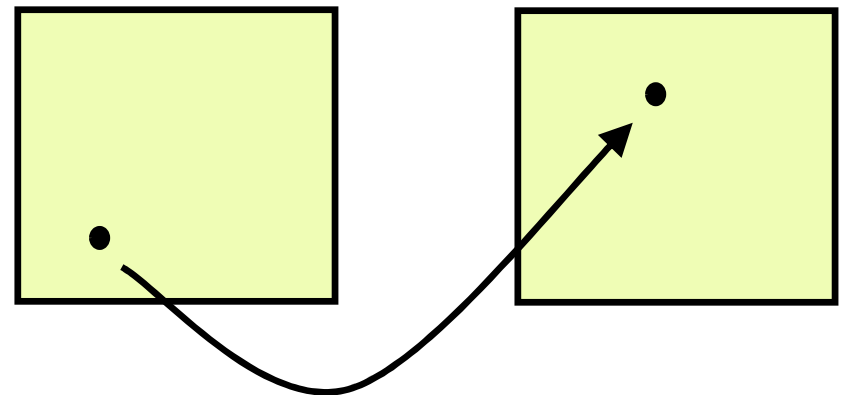
Geometric Operations

- Operation depend on ***pixel's coordinates***.
- Independent of pixels value.
- Context free.
- Example: translation

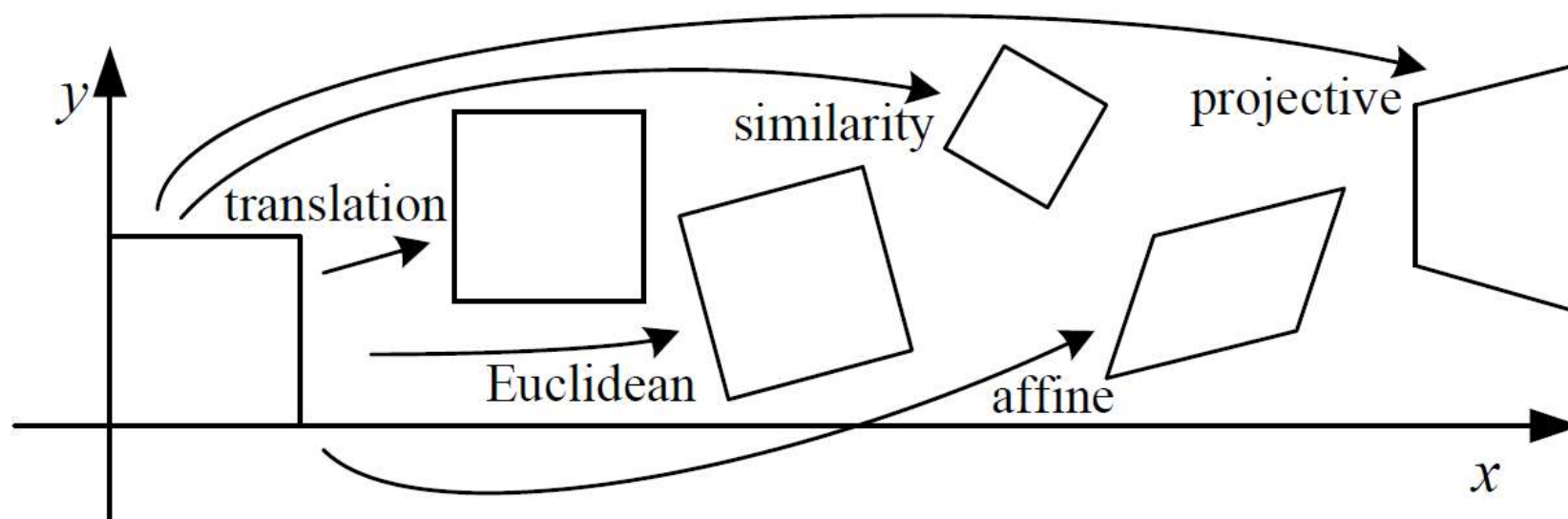


$$g(x, y) = f(G(x, y))$$

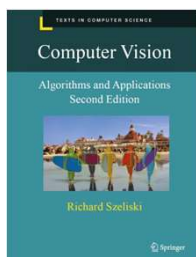
$$g(x, y) = f(x + a, y + b)$$



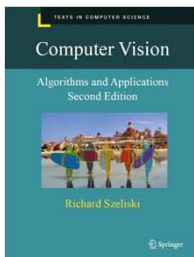
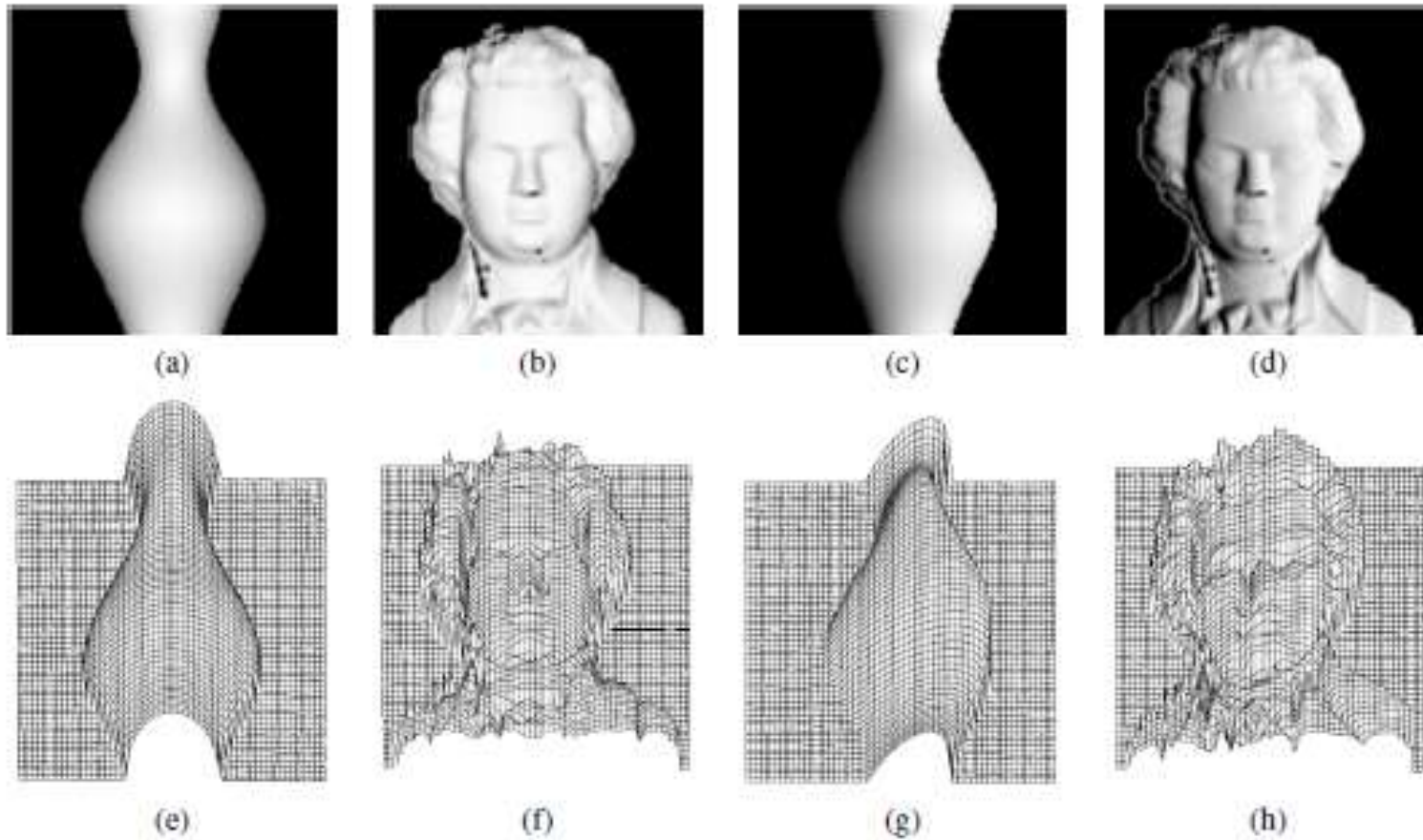
Geometric Operations



Transformation	Matrix	# DoF	Preserves	Icon
translation	$\begin{bmatrix} \mathbf{I} & \mathbf{t} \end{bmatrix}_{3 \times 4}$	3	orientation	
rigid (Euclidean)	$\begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix}_{3 \times 4}$	6	lengths	
similarity	$\begin{bmatrix} s\mathbf{R} & \mathbf{t} \end{bmatrix}_{3 \times 4}$	7	angles	
affine	$\begin{bmatrix} \mathbf{A} \end{bmatrix}_{3 \times 4}$	12	parallelism	
projective	$\begin{bmatrix} \tilde{\mathbf{H}} \end{bmatrix}_{4 \times 4}$	15	straight lines	



Shape from shading



Human stereo from single image?



Two view stereo



Key Idea: use feature motion to understand shape

Photometric Stereo



Key Idea: use pixel brightness to understand shape

Reflectance map

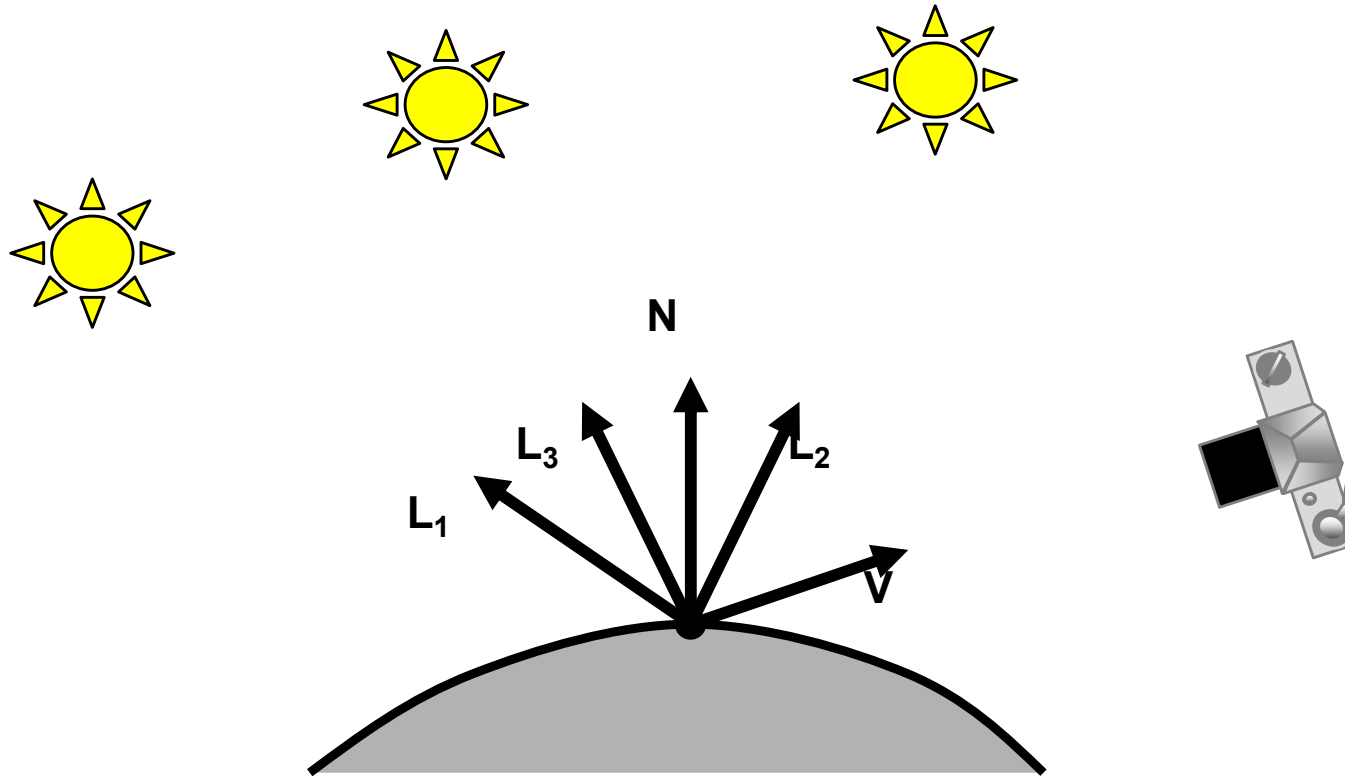


Assumption: uniform albedo

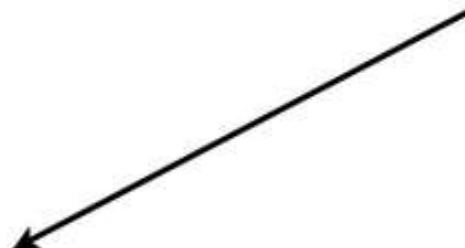
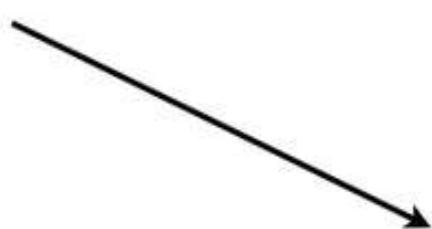


[Antiquity in colour](#) – Gallo-Roman Museum - Tongeren

Specular images: multiple lightsources

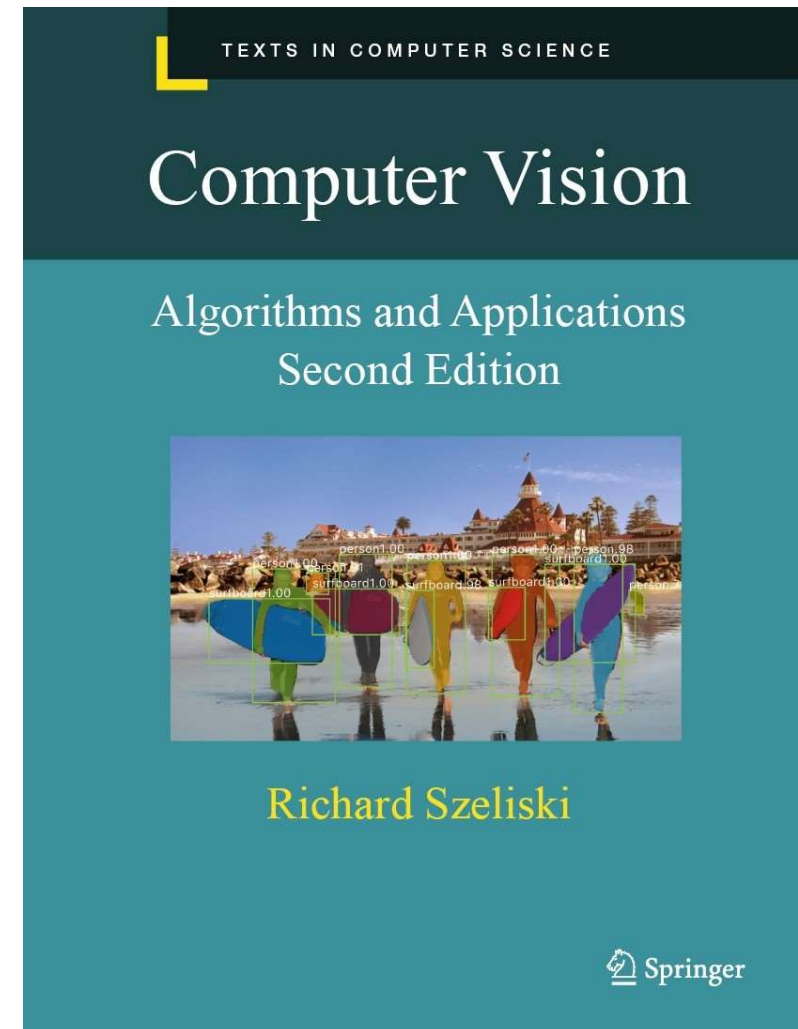


Each light-source has its own Reflectance map $R(p,q)$



Textbook

- ~~13.1.1~~
- 3.2
- 3.3
- 3.4



Neighborhood Operations

- Operation depends on ***Pixel's value*** and its spatial ***neighbors***.

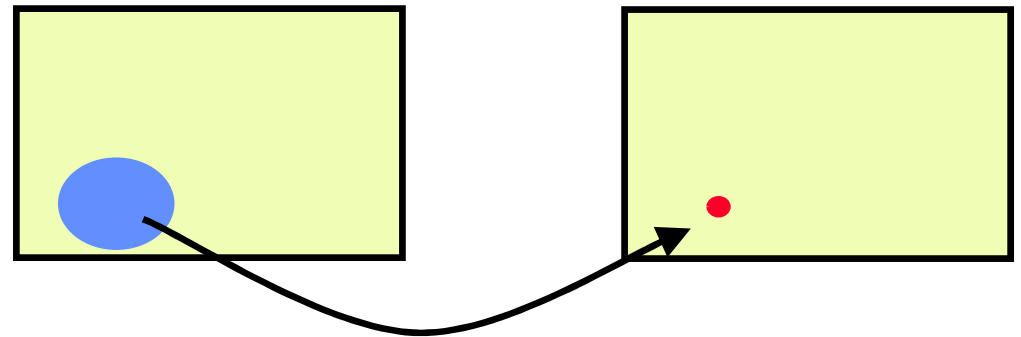
- Context dependent.

- Example:

$$g(x, y) = \sum_{i, j \in N(x, y)} f(i, j) / n$$



$$g(x, y) = M(\{f(i, j) | (i, j) \in N(x, y)\})$$



Outline

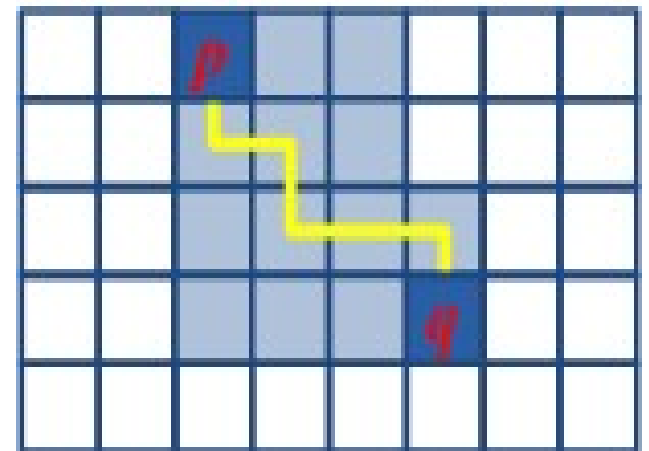
- Neighborhood image processing,
 - Connected Component Analysis (CCA)
 - Mathematical Morphology
 - Dilation
 - Corrosion
 - Opening
 - Closing
- Image filtering
 - Convolution
 - Frequently used filters
 - Gaussian filter
 - Nonlinear Filters
 - Median filter
 - Bilateral filter
- Frequency domain analysis

Connected Components Analysis

- Description
 - Connected Components Analysis is a multiple-class labeling operation.
 - It performs the change from **pixels** to **regions**.
 - All pixels in the same region are given the same label.

Connected Component Operators

- Definition of Connected Component
 - Two pixels p and q belong to the same connected component C if there is a sequence of 1-pixels $(p_0, p_1, \dots, p_n)$, where
 - $p_0 = p$
 - $p_n = q$
 - $p_{i-1}, p_i : i = 1, \dots, n$ are neighbors



Neighborhood Types

		•		
	•	x	•	
		•		

4-connected

	•	•	•	
	•	x	•	
	•	•	•	

8-connected

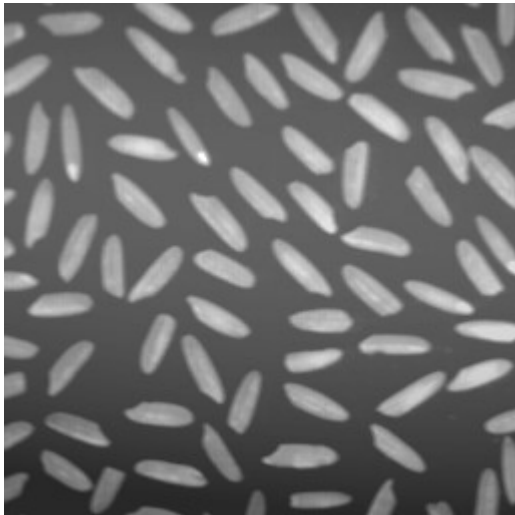
0	1	1	0	1	0	0
0	1	1	0	1	0	1
1	1	1	0	1	0	1
0	0	0	0	1	1	1
0	1	0	0	0	0	0
0	1	1	1	1	1	0
0	1	1	1	0	0	0

Original Image

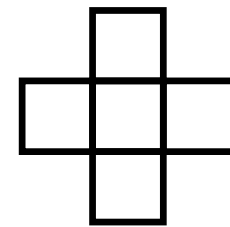
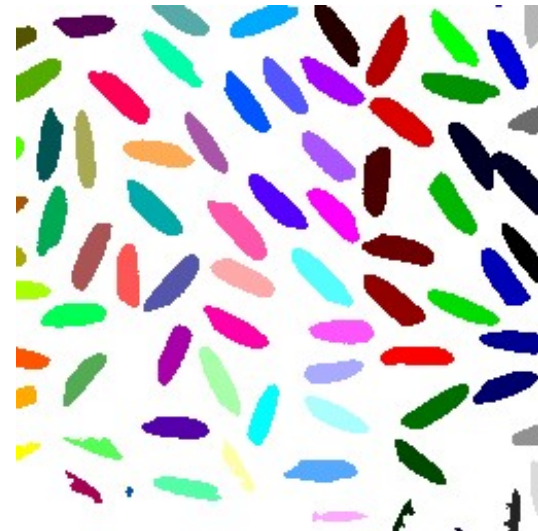
0	1	1	0	2	0	0
0	1	1	0	2	0	2
1	1	1	0	2	0	2
0	0	0	0	2	2	2
0	3	0	0	0	0	0
0	3	3	3	3	3	0
0	3	3	3	0	0	0

Connected Components

An application example: Rice grain counting and measuring



Histogram Thresholding →



4-way connected

Another application example

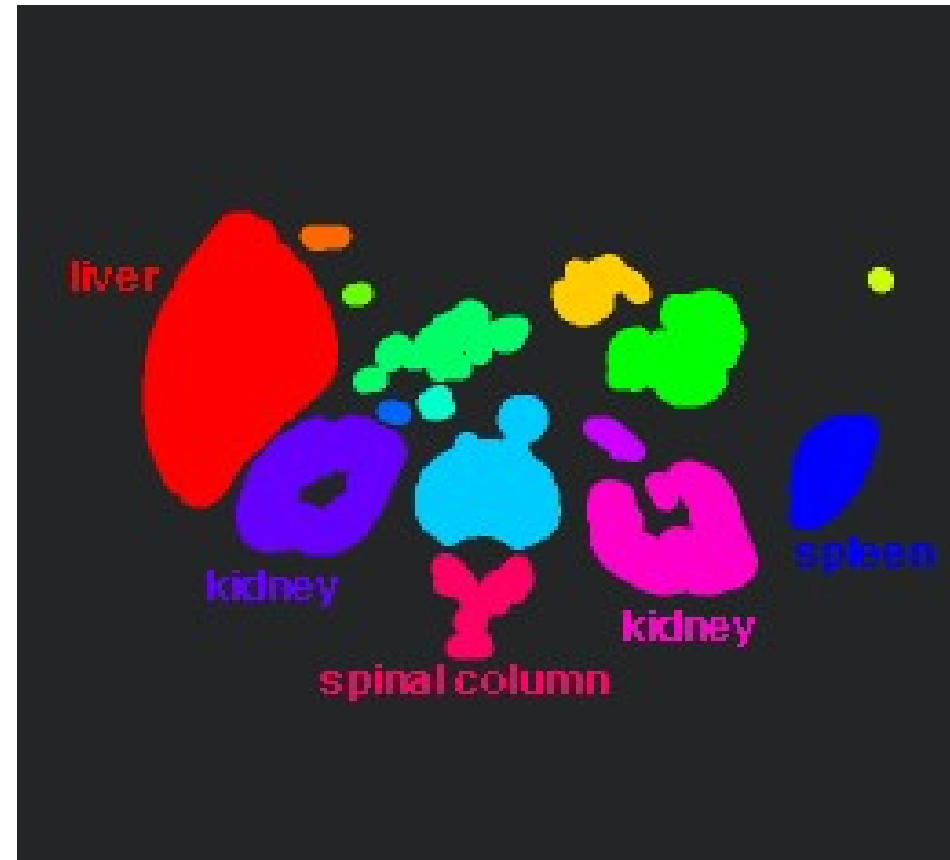
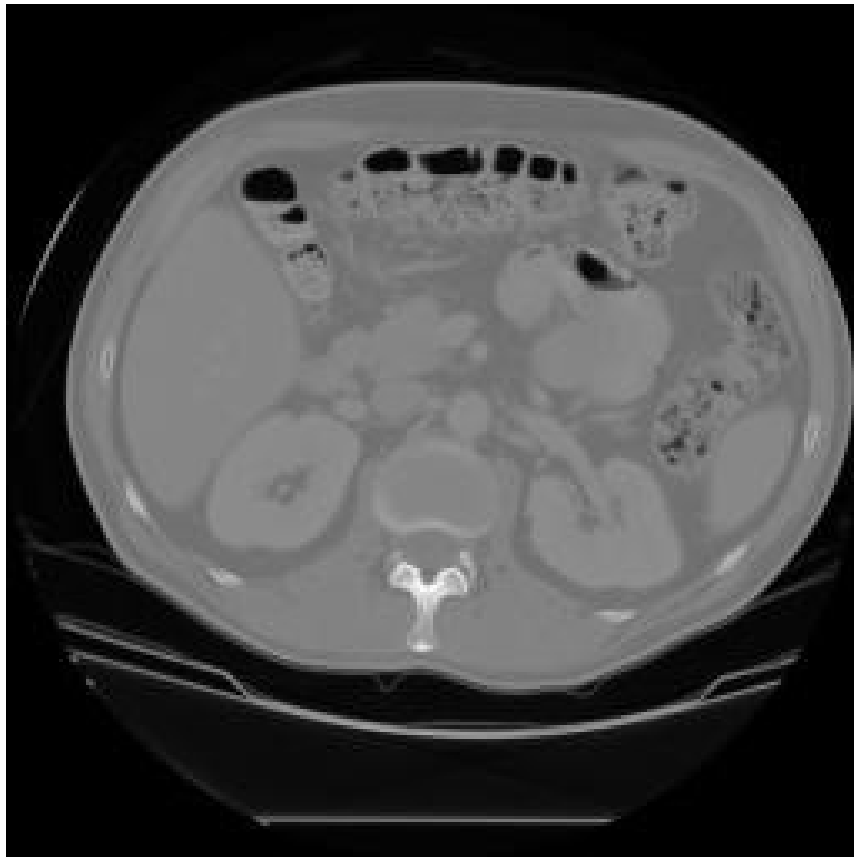


original gray tone image



binary thresholded image

Labeling shown as Pseudo-Color

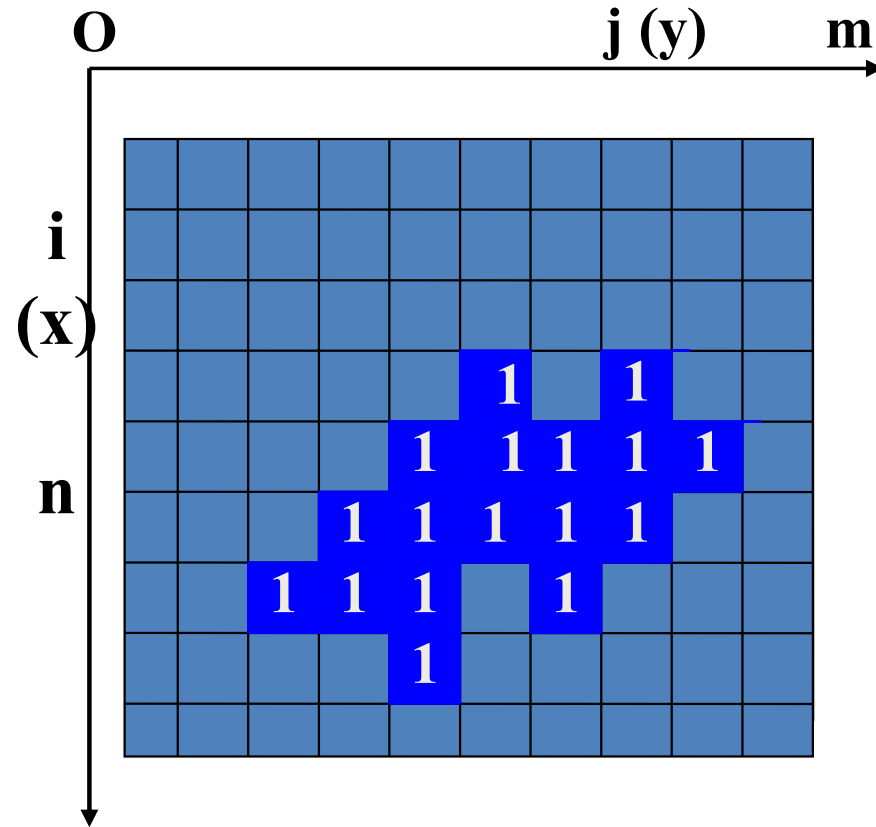


Region property computation

- For each connected region, we can compute many of its properties, e.g.
 - Areas
 - Perimeter
 - Centre of gravity
 - circularity,
 - Major axis
 - Minor axis
 - mean and standard deviation of radial distance
 - bounding box
 - extremal axis length from bounding box
 - second order moments (row, column, mixed)
 - ...

Explain this using an example

b_{ij} : value at pixel in row i & column j .



Explain this using an example

b_{ij} : value at pixel in row i & column j .

Assume pixel area is 1.

Area:
$$A = \sum_{i=1}^n \sum_{j=1}^m b_{ij}$$

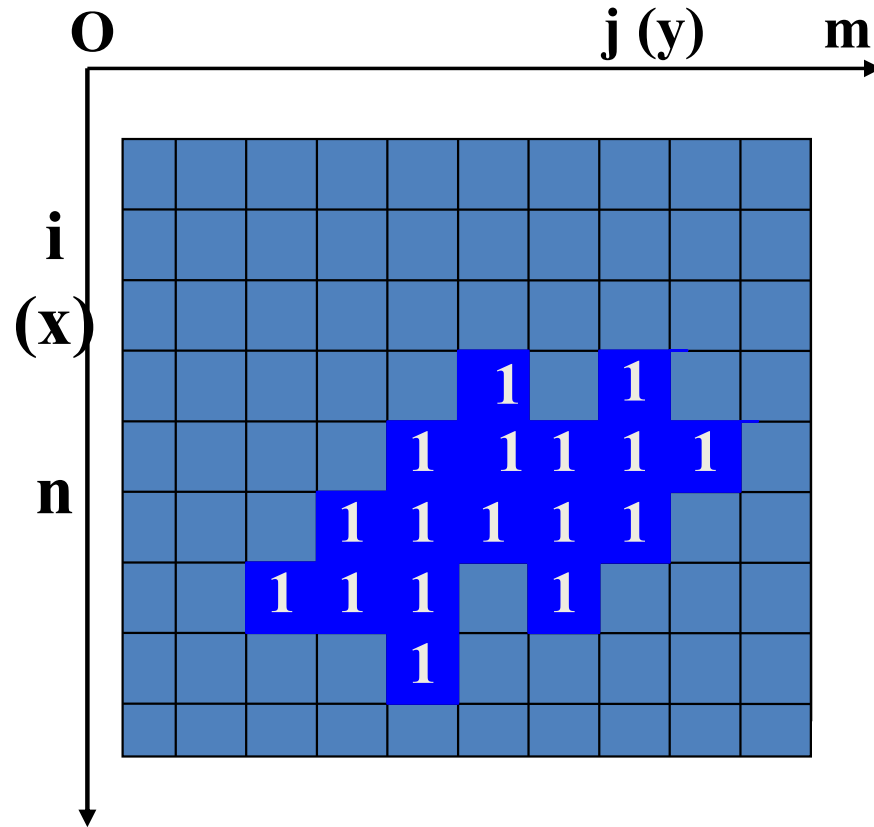
Center of Mass:

$$\bar{x} = (1 / A) \sum_{i=1}^n \sum_{j=1}^m i b_{ij}$$

$$\bar{y} = (1 / A) \sum_{i=1}^n \sum_{j=1}^m j b_{ij}$$

Second Moments.

$$a' = \sum_{i=1}^n \sum_{j=1}^m i^2 b_{ij}, \quad b' = 2 \sum_{i=1}^n \sum_{j=1}^m ij b_{ij}, \quad c' = \sum_{i=1}^n \sum_{j=1}^m j^2 b_{ij}$$



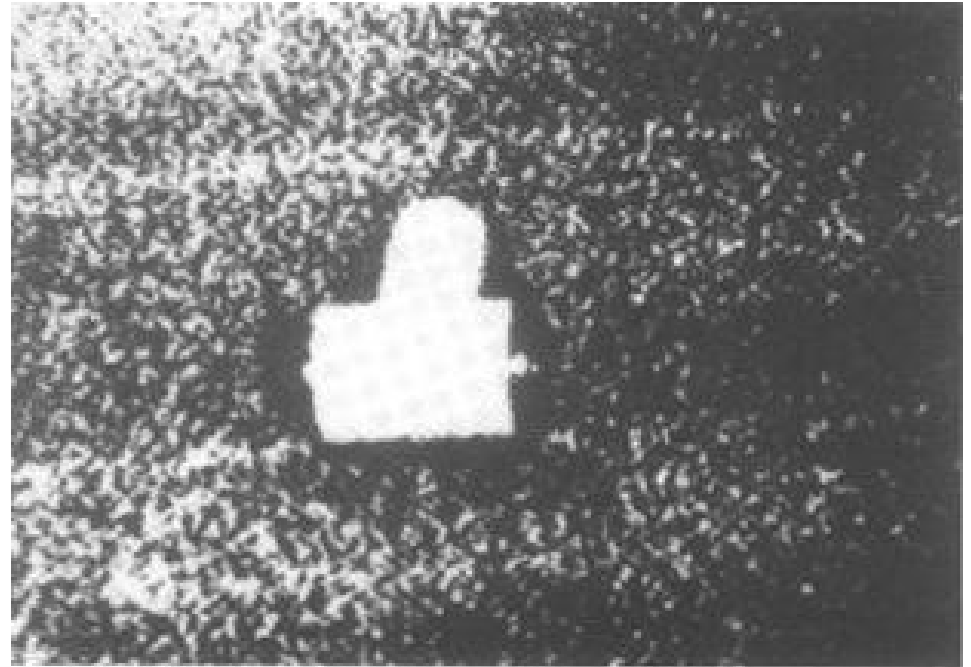
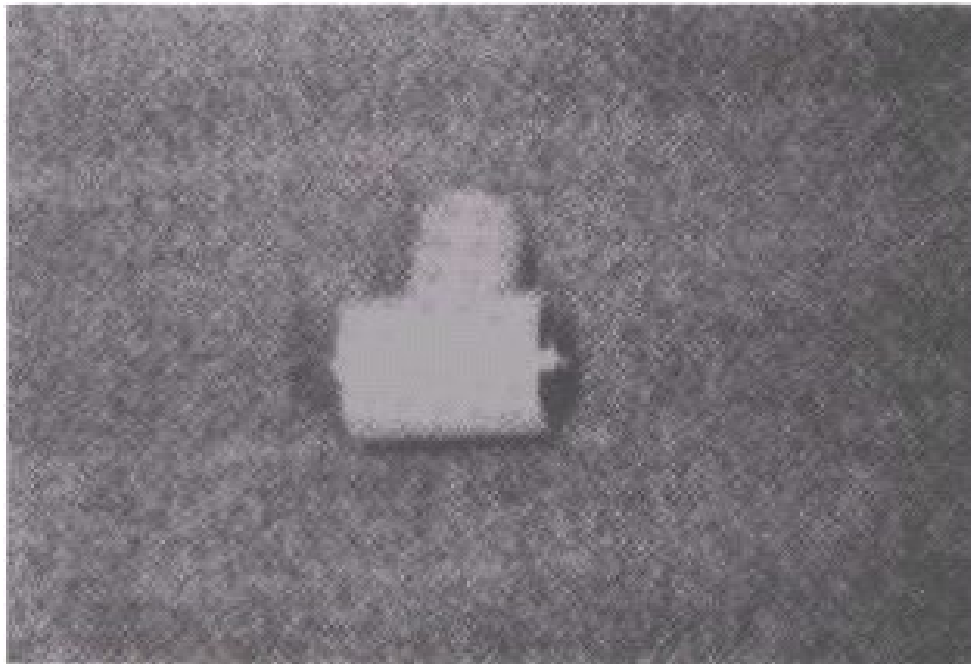
Outline

- Neighborhood image processing
 - Connected Component Analysis (CCA)
 - Mathematical Morphology
 - Dilation
 - Corrosion
 - Opening
 - Closing
- Image filtering
 - Convolution
 - Frequently used filters
 - Gaussian filter
 - Nonlinear Filters
 - Median filter
 - Bilateral filter
- Frequency domain analysis

- **Morphology:** originally, a branch of biology science that studies the form and structure of animals and plants
- Mathematical Morphology in image processing is used to extract image components for representation and description of region shape, such as boundaries, skeletons, and the convex hull.

Motivation: remove small noise regions

- Example result by a binarization algorithm



Mathematical Morphology

In binary image processing, mathematical morphology consists of two basic operations,

dilation , erosion

and several composite operations

closing , opening, . . .

Outline

- Neighborhood image processing
 - Connected Component Analysis (CCA)
 - Mathematical Morphology
 - Dilation
 - Erosion
 - Opening
 - Closing
- Image filtering
 - Convolution
 - Frequently used filters
 - Gaussian filter
 - Nonlinear Filters
 - Median filter
 - Bilateral filter
- Frequency domain analysis
 - 2D FFT

Structuring Elements

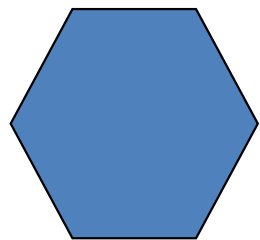
A **structuring element** is a shape mask used in the basic morphological operations.

They can be any shape and size that is digitally representable, and each has an **origin**.

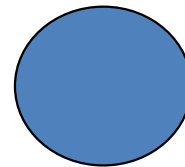


box

`box(length, width)`

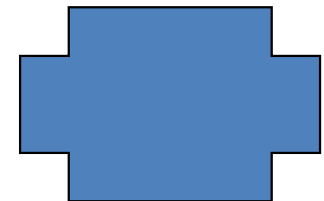


hexagon



disk

`disk(diameter)`



something

DILATION

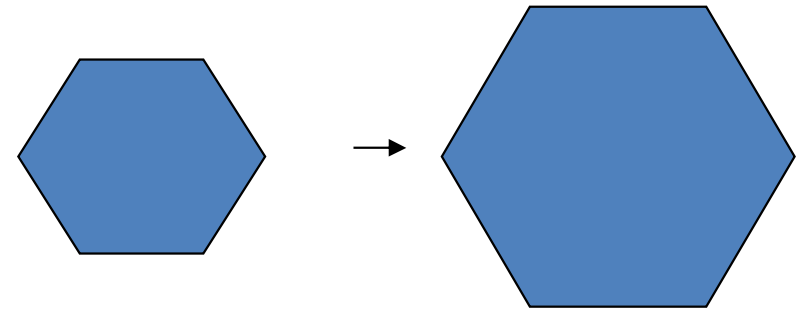
`dilate(B,S)` takes binary image B , places the origin of structuring element S over each 1-pixel, and ORs the structuring element S into the output image at the corresponding position.

Dilation

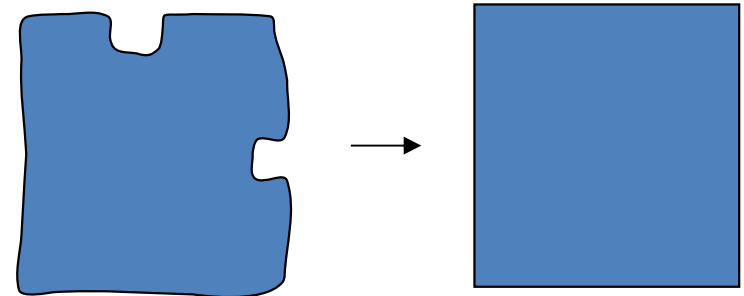
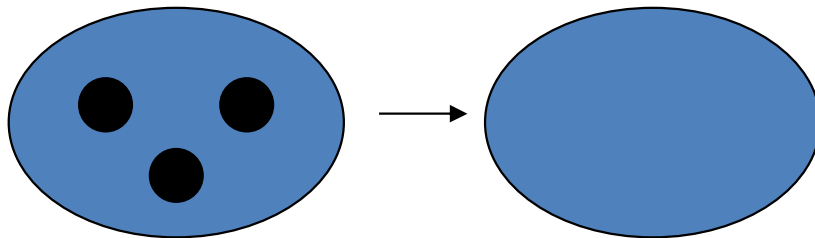
Dilation **expands** the connected sets of 1s of a binary image.

It can be used for

1. growing features



2. filling holes and gaps



Example for Dilation

Input image

0	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---

Step 1 Point-wise 'AND'

Structuring Element

1	1	1
---	---	---

Step 2 'OR' over the structure

Output Image

	0								
--	---	--	--	--	--	--	--	--	--

Example for Dilation

Input image

0	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0							
--	---	---	--	--	--	--	--	--	--

Example for Dilation

Input image

0	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	1						
--	---	---	---	--	--	--	--	--	--

Example for Dilation

Input image

0	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	1	1					
--	---	---	---	---	--	--	--	--	--

Example for Dilation

Input image

0	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	1	1	1				
--	---	---	---	---	---	--	--	--	--

Example for Dilation

Input image

0	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	1	1	1	1			
--	---	---	---	---	---	---	--	--	--

Example for Dilation

Input image

0	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	1	1	1	1	1		
--	---	---	---	---	---	---	---	--	--

Example for Dilation

Input image

0	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	1	1	1	1	1	1	
--	---	---	---	---	---	---	---	---	--

Example for Dilation

Input image

0	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

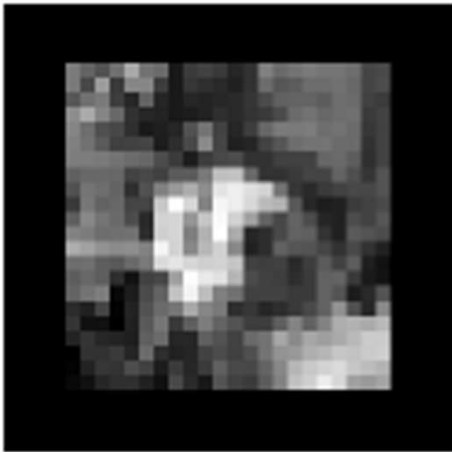
1	1	1
---	---	---



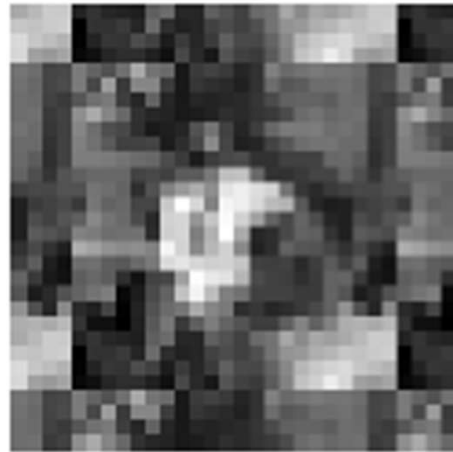
Output Image

?	0	0	1	1	1	1	1	1	?
---	---	---	---	---	---	---	---	---	---

How to handle borders?



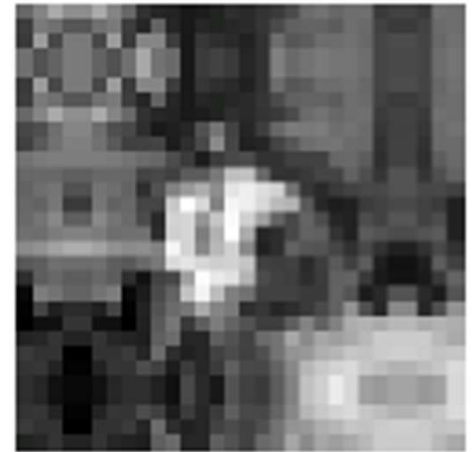
zero



wrap

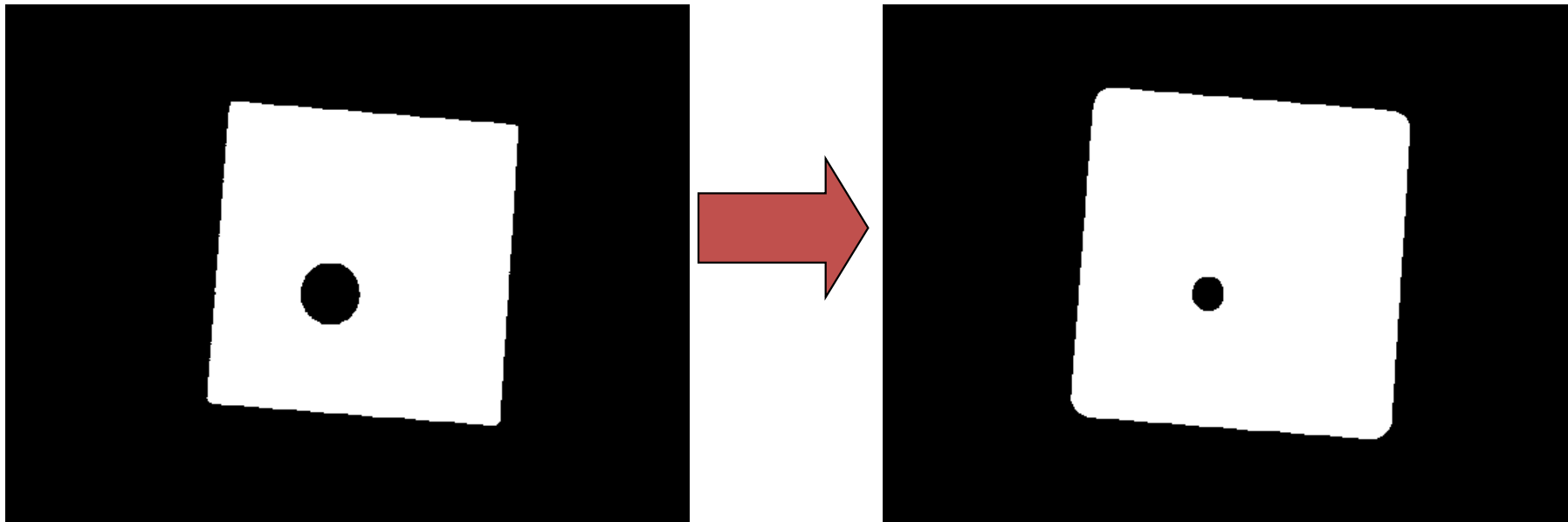


clamp



mirror

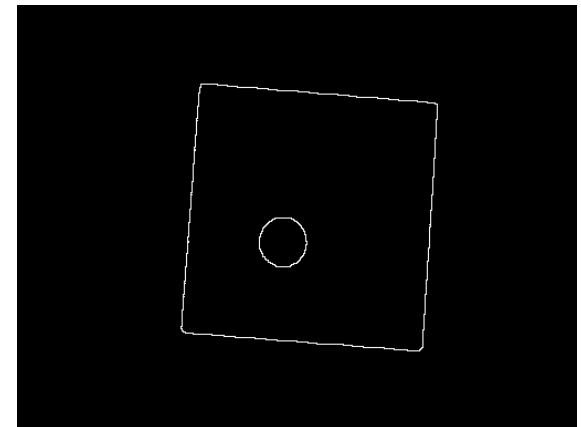
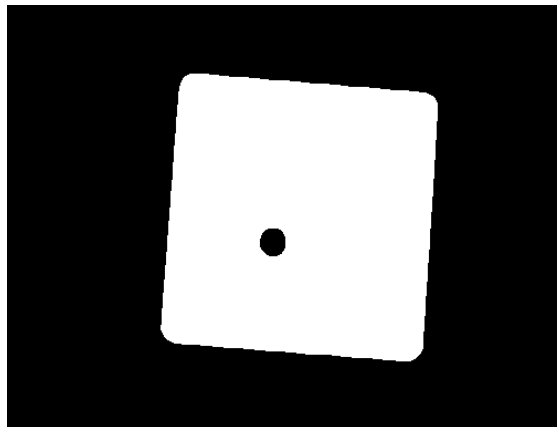
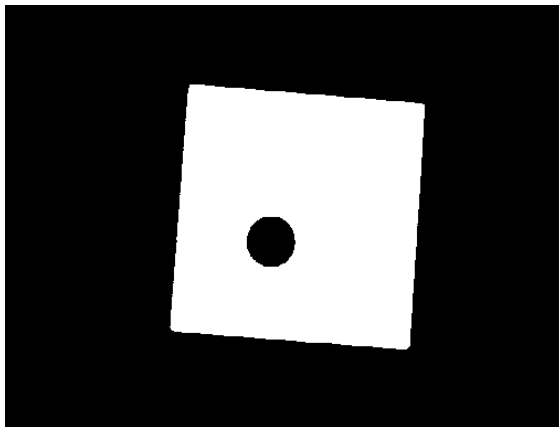
Another Dilation Example



- Image get lighter, more uniform intensity

Edge detection example

- Edge Detection
 1. Dilate input image
 2. Subtract input image from dilated image
 3. Edges remain!



Outline

- Neighborhood image processing
 - Connected Component Analysis (CCA)
 - Mathematical Morphology
 - Dilation
 - Erosion
 - Opening
 - Closing
- Image filtering
 - Convolution
 - Frequently used filters
 - Gaussian filter
 - Nonlinear Filters
 - Median filter
 - Bilateral filter
- Frequency domain analysis

EROSION

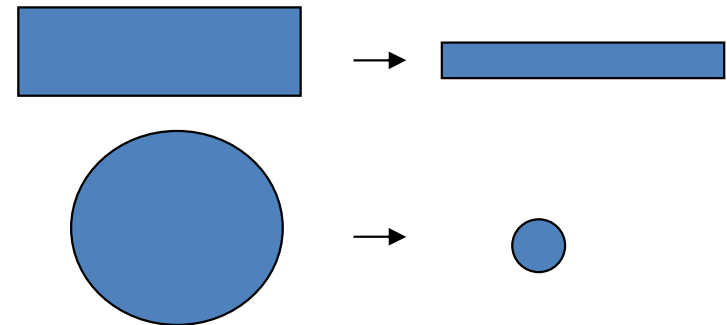
`erode(B,S)` takes a binary image B , places the origin of structuring element S over every pixel position, and AND a binary 1 into that position of the output image, if and only if every position of S (with a 1) covers a 1 in B .

Erosion

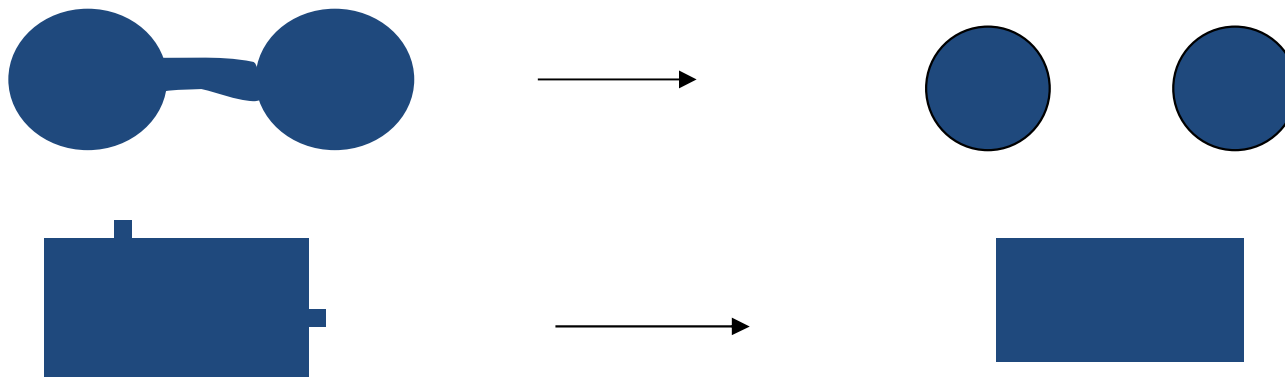
Erosion **shrinks** the connected sets of 1s of a binary image.

It can be used for

1. shrinking features

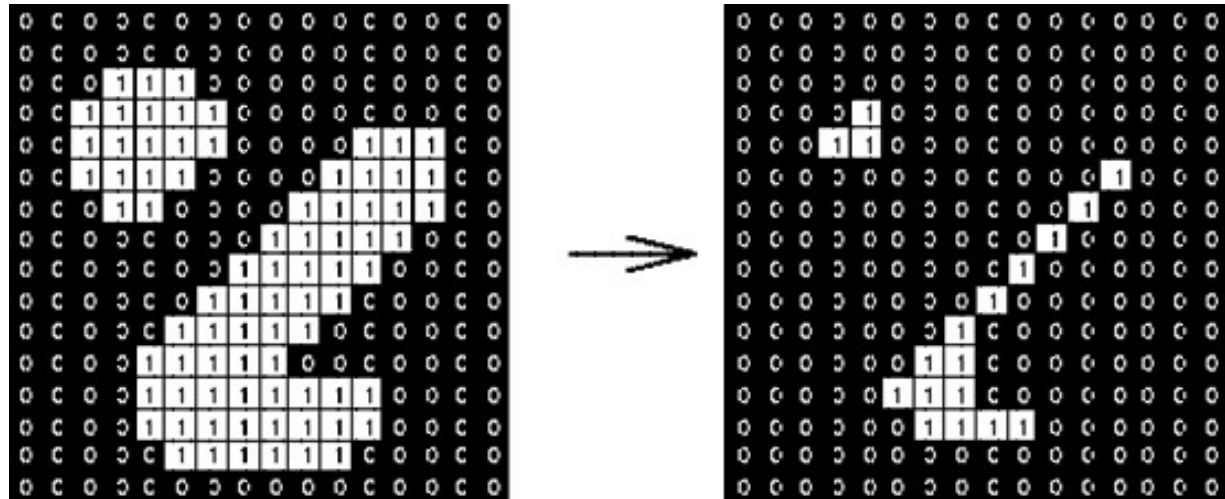


2. Removing bridges, branches and small protrusions



Erosion

- **Erosion** is a basic morphological operation



- Applied **Structuring Element**:

1	1	1
1	1	1
1	1	1

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---

Step 1 Point-wise 'AND'

Structuring Element

1	1	1
---	---	---

Step 2 'AND' over the structure

Output Image

	0								
--	---	--	--	--	--	--	--	--	--

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0							
--	---	---	--	--	--	--	--	--	--

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	0						
--	---	---	---	--	--	--	--	--	--

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	0	0					
--	---	---	---	---	--	--	--	--	--

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	0	0	1				
--	---	---	---	---	---	--	--	--	--

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	0	0	1	0			
--	---	---	---	---	---	---	--	--	--

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	0	0	1	0	0		
--	---	---	---	---	---	---	---	--	--

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

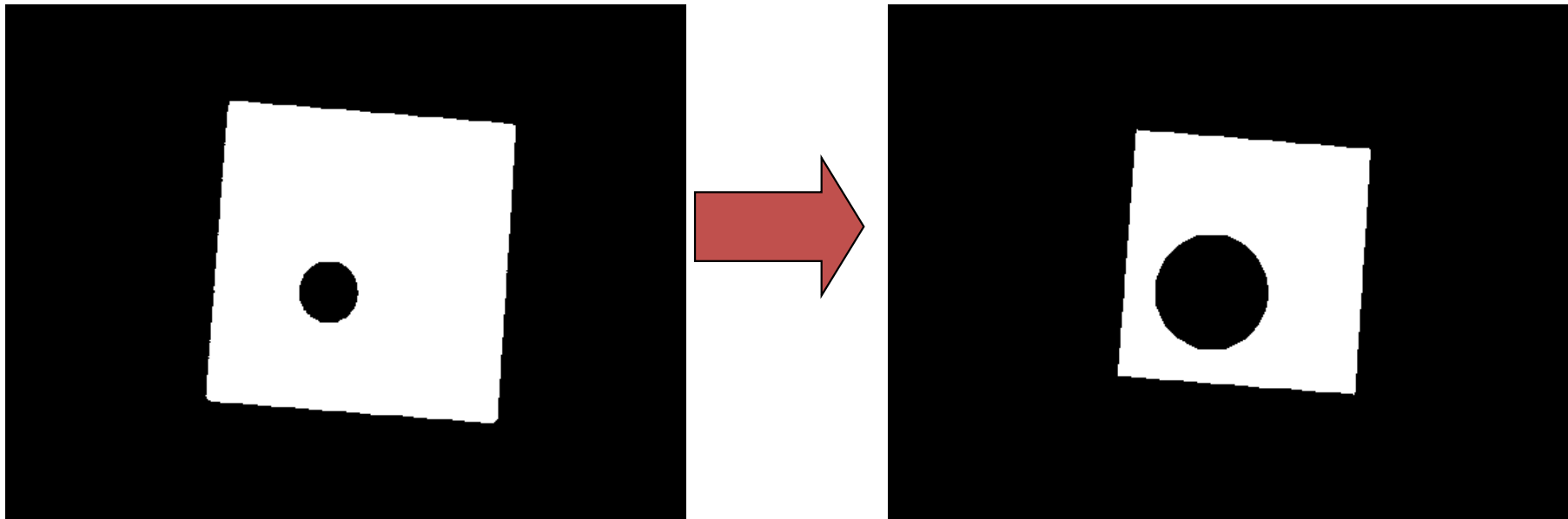
1	1	1
---	---	---



Output Image

	0	0	0	0	1	0	0	0	
--	---	---	---	---	---	---	---	---	--

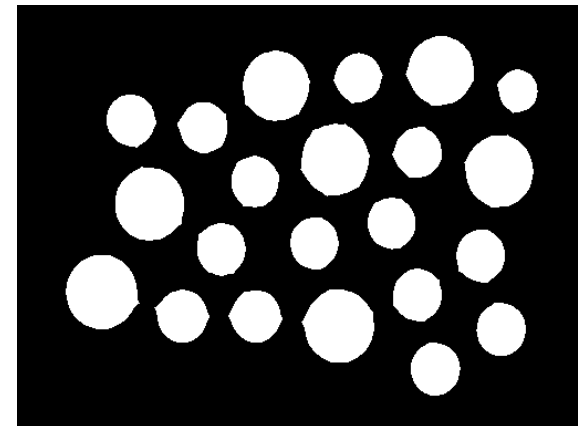
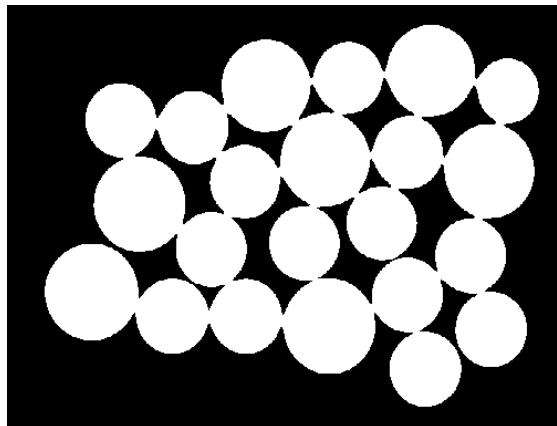
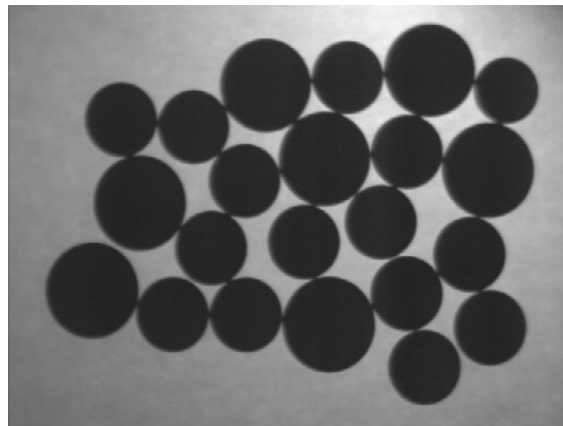
An example of erosion



- White = 0, black = 1, dual property, image as a result of erosion gets darker

Counting Coins

- Counting coins is difficult because they touch each other!
- Solution: Binarization and Erosion separates them!



Outline

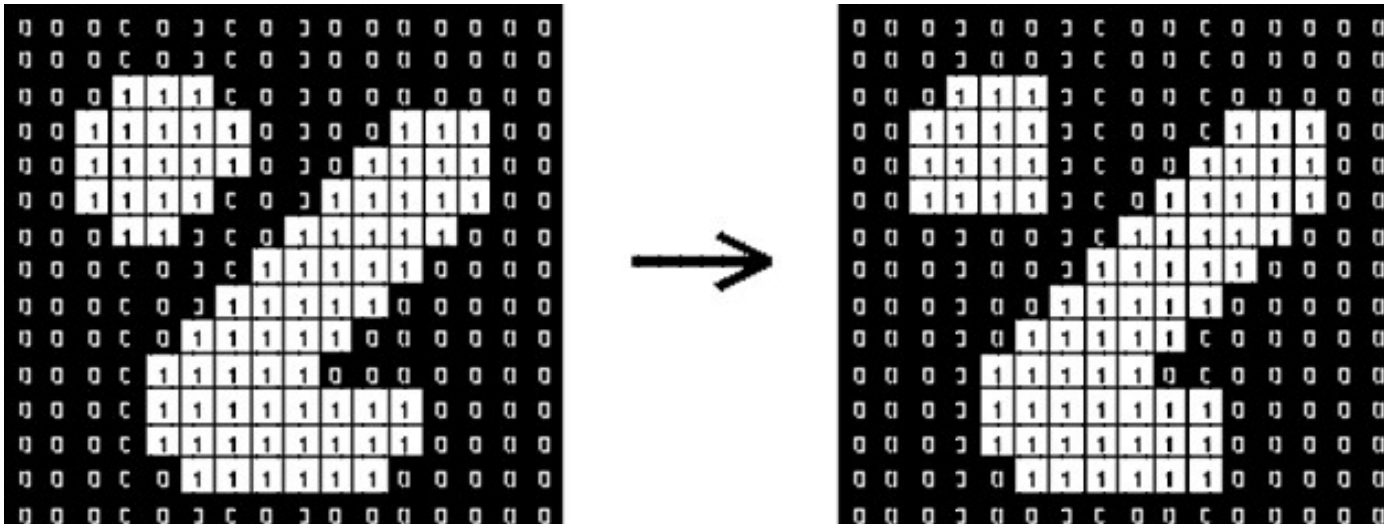
- Neighborhood image processing
 - Connected Component Analysis (CCA)
 - Mathematical Morphology
 - Dilation
 - Corrosion
 - Opening
 - Closing
- Image filtering
 - Convolution
 - Frequently used filters
 - Gaussian filter
 - Nonlinear Filters
 - Median filter
 - Bilateral filter
- Frequency domain analysis

Opening

- Similar to Erosion
 - Spot and noise removal
 - Less destructive
- Erosion next dilation
- *the same structuring element for both operations.*

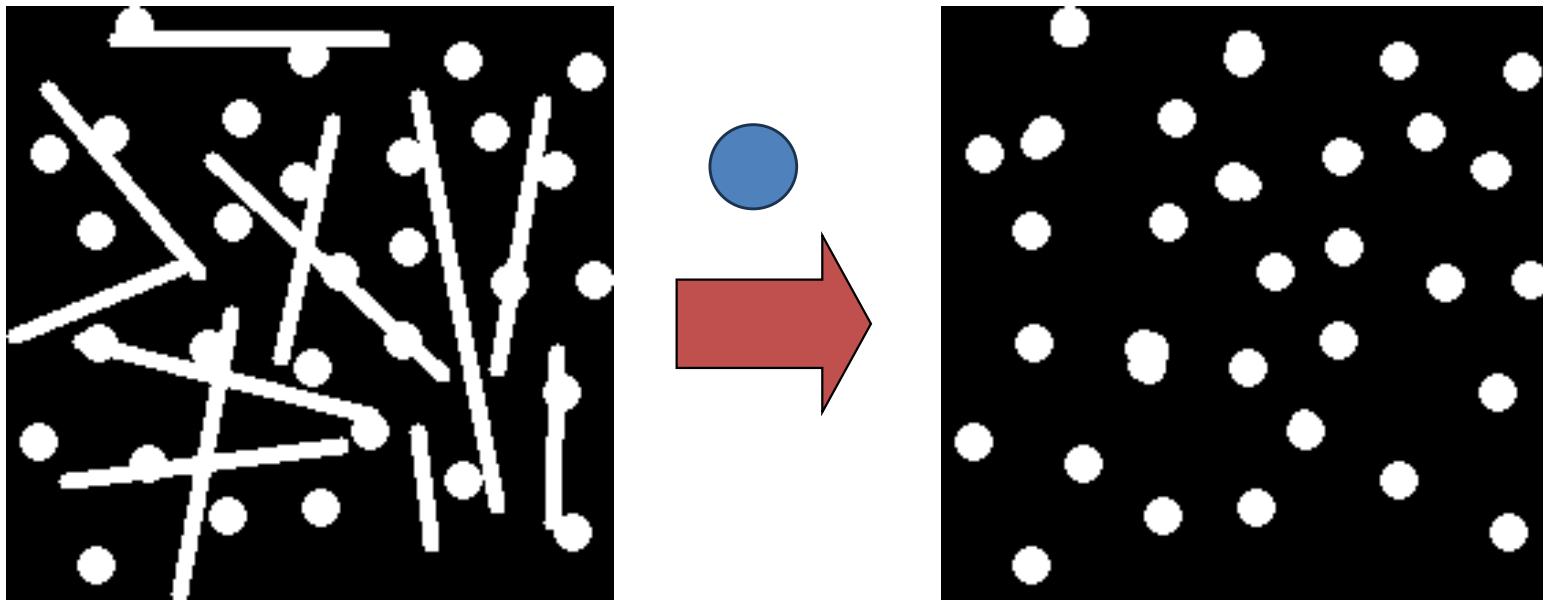
Opening

- Structuring element: 3x3 square

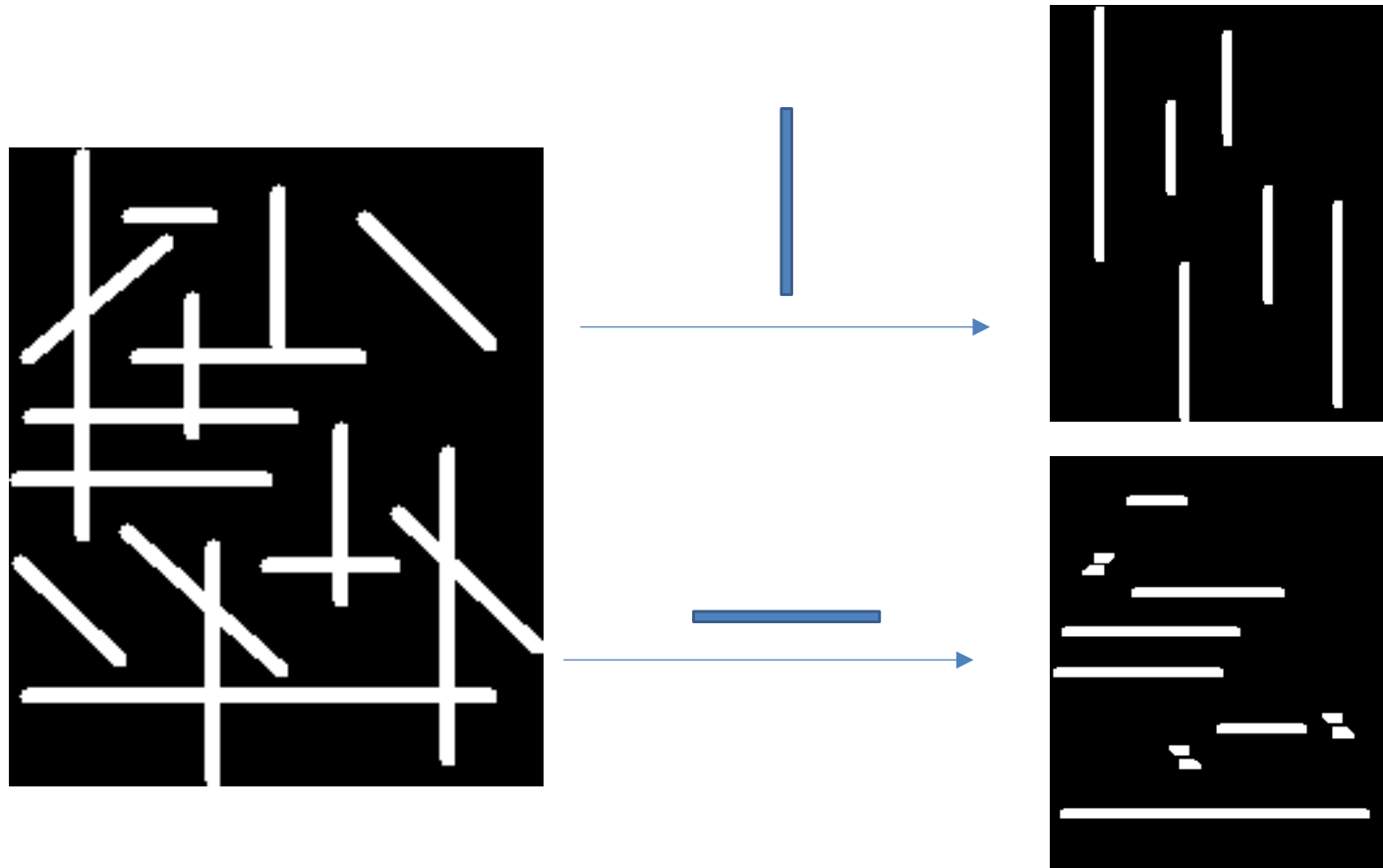


Opening Example

- Opening with a 11 pixel diameter disc
- Erosion plus dilation

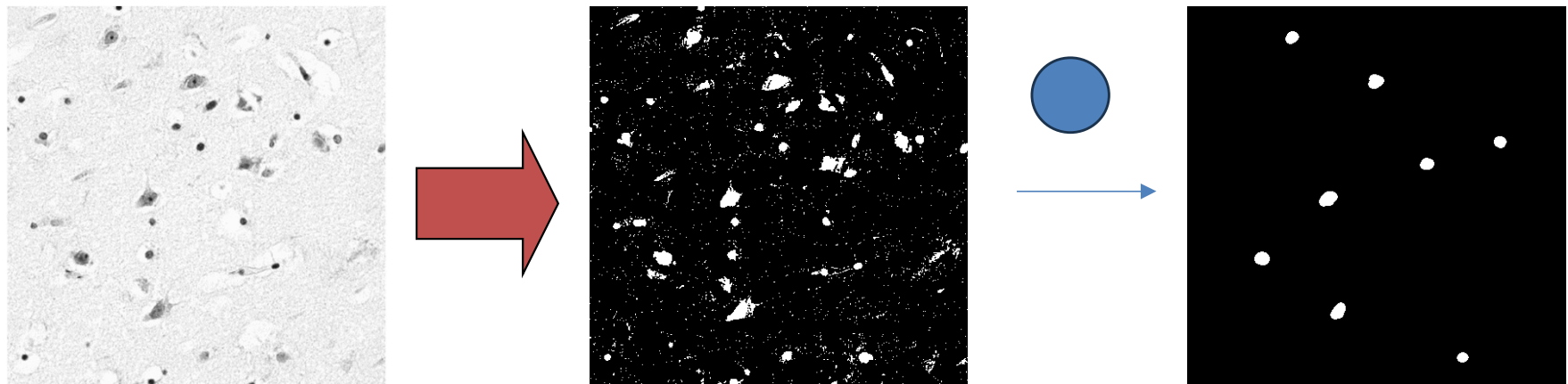


Opening Example



Use Opening for Separating Blobs

- Use large structuring element that fits into the big blobs
- Structuring Element: disc



Outline

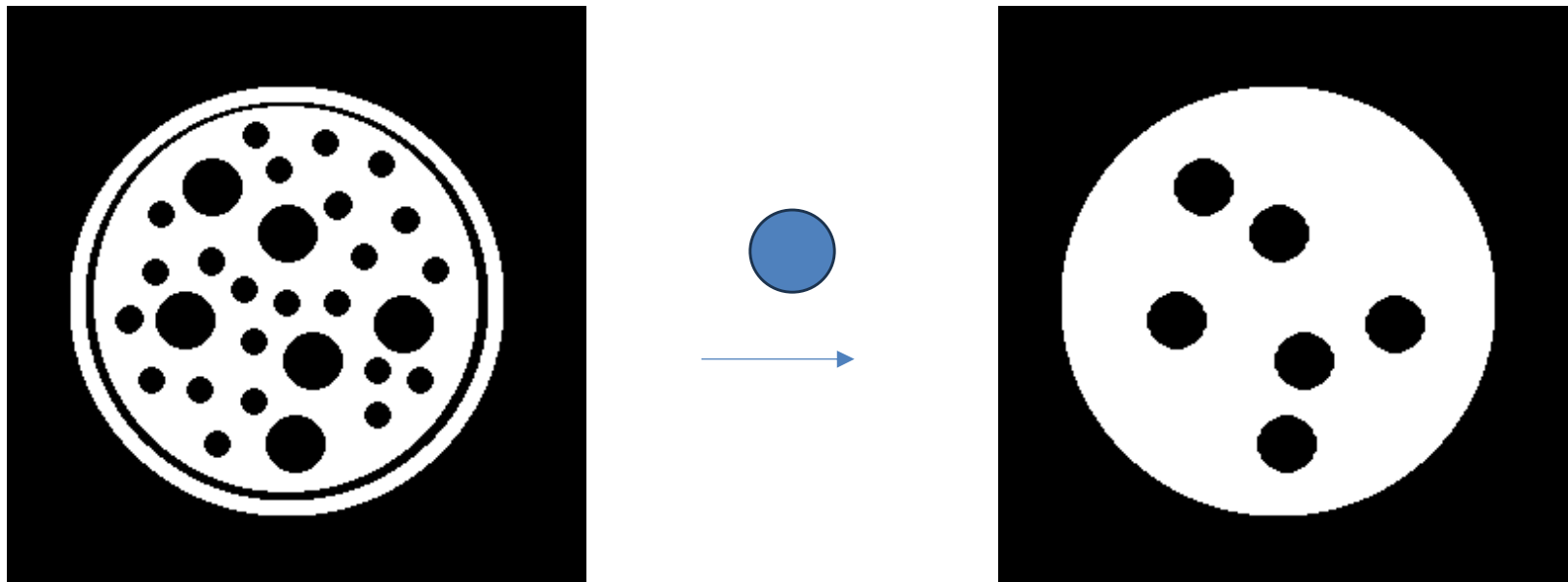
- Neighborhood image processing
 - Connected Component Analysis (CCA)
 - Mathematical Morphology
 - Dilation
 - Corrosion
 - Opening
 - Closing
- Image filtering
 - Convolution
 - Frequently used filters
 - Gaussian filter
 - Nonlinear Filters
 - Median filter
 - Bilateral filter
- Frequency domain analysis

Closing

- Similar to Dilation
 - Removal of holes
 - Tends to enlarge regions, shrink background
- Dilation next erosion.

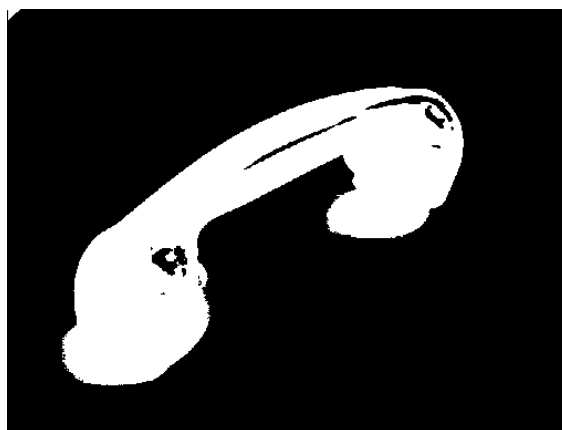
Closing Example

- Closing operation with a 22 pixel disc
- Closes small holes in the foreground



Closing Example

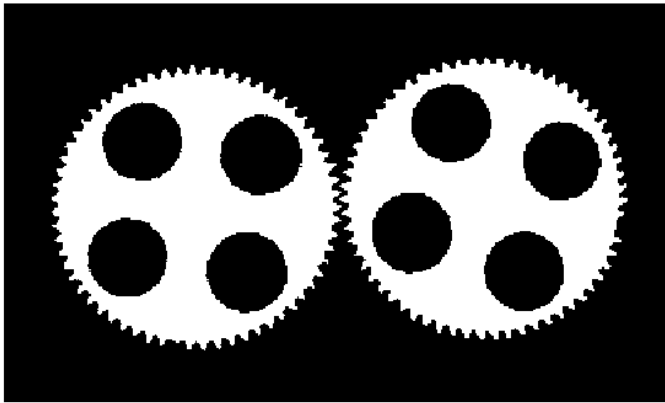
1. Threshold
2. Closing with disc of size 20



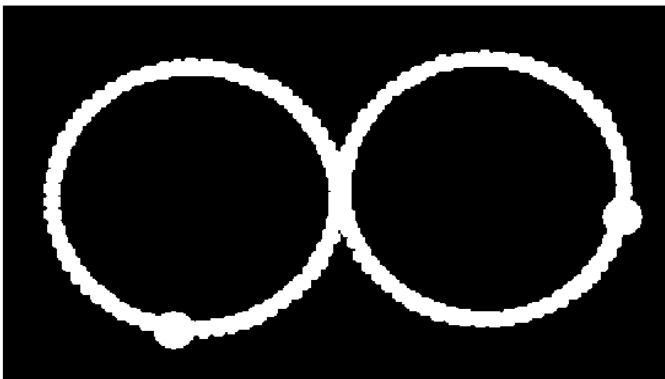
Opening & Closing

- Derived from the two basic operations
 - Dilation
 - Erosion
- Opening and closing are dual operations
 - *i.e.* opening the foreground pixels with a particular structuring element
 - is equivalent to closing the background pixels with the same element.

For your own reading:
Sample Application: Gear-tooth Inspection



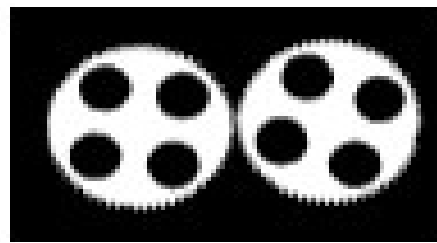
original
binary
image



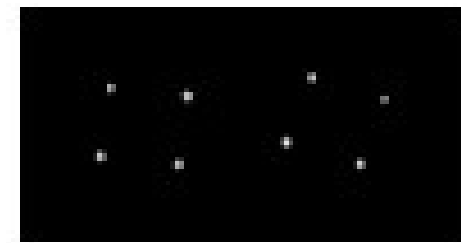
detected
defects

One solution

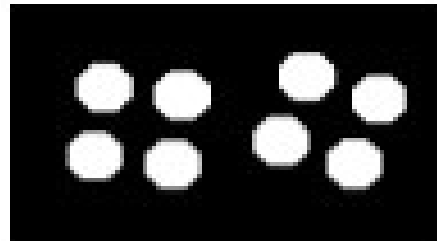
For fun!



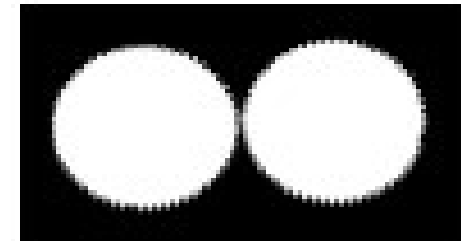
a) original image B



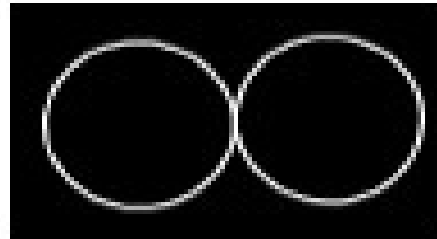
b) $B1 = B \oplus \text{hole_mask}$



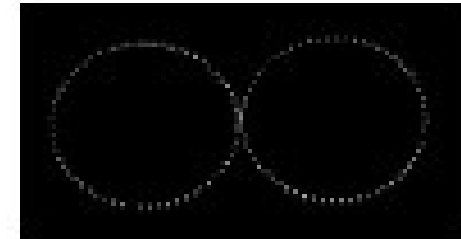
c) $B2 = B1 \oplus \text{hole_mask}$



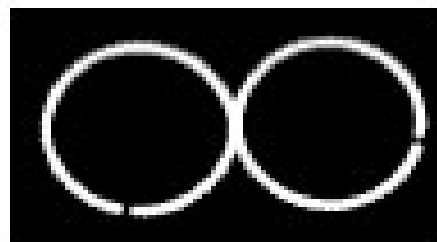
d) $B3 = B \text{ OR } B2$



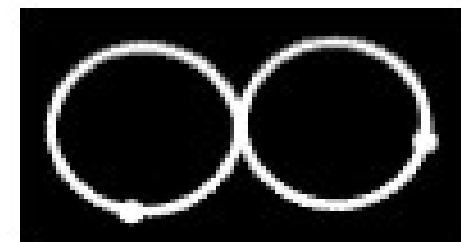
e) B7 (see text)



f) $B8 = B \text{ AND } B7$



g) $B9 = B8 \oplus \text{tip_spacing}$



h) $\text{RESULT} = ((B7 - B9) \oplus \text{defect_mask}) \text{ OR } B9$

Outline

- Neighborhood image processing,
 - Connected Component Analysis (CCA)
 - Mathematical Morphology
 - Dilation
 - Corrosion
 - Opening
 - Closing
- Image filtering
 - Convolution
 - Frequently used filters
 - Gaussian filter
 - Nonlinear Filters
 - Median filter
 - Bilateral filter
- Frequency domain analysis

Motivation: Noise reduction

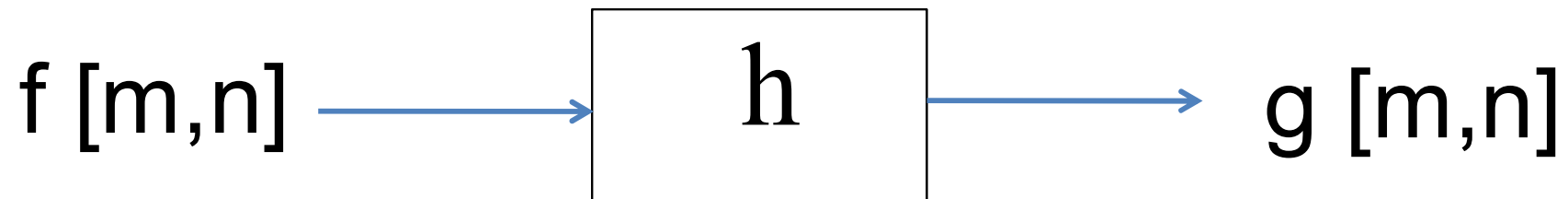
- Given a camera and a still scene, how can you reduce noise?



Filtering

Input image

Output image



filter, kernel,
impulse
response

Example Moving average

- Let's replace each pixel with a *weighted* average of its neighborhood
- The neighborhood mark is called a filter or a *kernel*
- What are the weights for the average of a 3x3 neighborhood?

$$\frac{1}{9}$$

1	1	1
1	1	1
1	1	1

“box filter”

Image filtering

$$h[\cdot, \cdot] = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$f[\cdot, \cdot]$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$g[\cdot, \cdot]$

$$g[m, n] = \sum_{k, l} h[k, l] f[m + k, n + l]$$

Image filtering

$$h[\cdot, \cdot] = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$f[\cdot, \cdot]$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$g[\cdot, \cdot]$

	0	10							

$$g[m, n] = \sum_{k, l} h[k, l] f[m + k, n + l]$$

Image filtering

$$h[\cdot, \cdot] = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$f[\cdot, \cdot]$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$g[\cdot, \cdot]$

	0	10	20						

$$g[m, n] = \sum_{k, l} h[k, l] f[m + k, n + l]$$

Image filtering

$$h[\cdot, \cdot] = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$f[\cdot, \cdot]$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$g[\cdot, \cdot]$

	0	10	20	30					

$$g[m, n] = \sum_{k, l} h[k, l] f[m + k, n + l]$$

Image filtering

$$h[\cdot, \cdot] = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$f[\cdot, \cdot]$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$g[\cdot, \cdot]$

	0	10	20	30	30				

$$g[m, n] = \sum_{k, l} h[k, l] f[m + k, n + l]$$

Image filtering

$$h[\cdot, \cdot] = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$f[\cdot, \cdot]$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$g[\cdot, \cdot]$

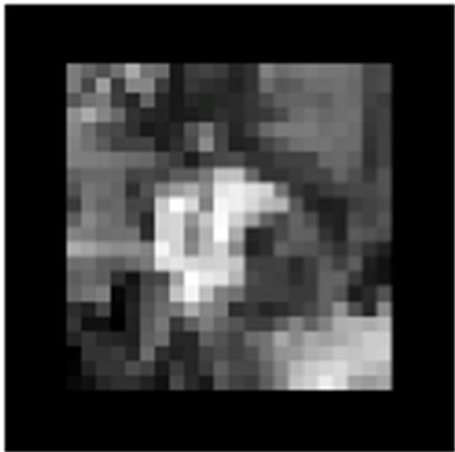
	0	10	20	30	30	30	20	10	
	0	20	40	60	60	60	40	20	
	0	30	60	90	90	90	60	30	
	0	30	50	80	80	90	60	30	
	0	30	50	80	80	90	60	30	
	0	20	30	50	50	60	40	20	
	10	20	30	30	30	30	20	10	
	10	10	10	0	0	0	0	0	

$$g[m, n] = \sum_{k, l} h[k, l] f[m + k, n + l]$$

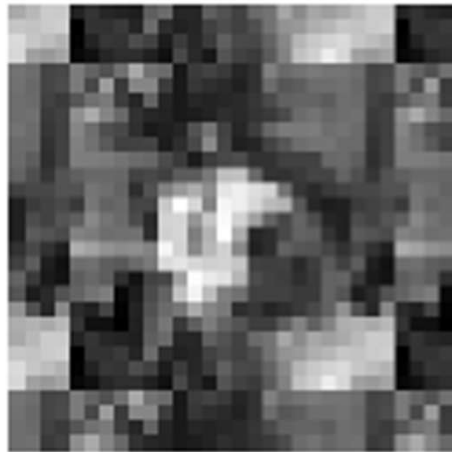
Smoothing with box filter



How to handle borders?



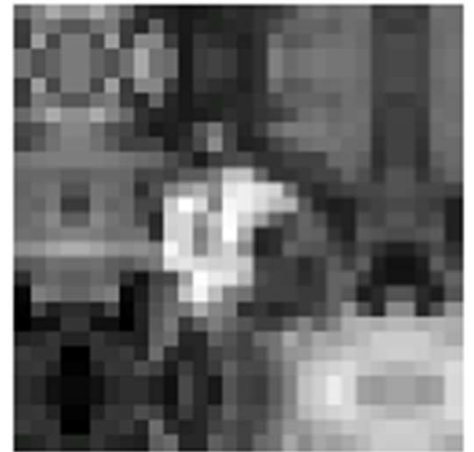
zero



wrap



clamp



mirror

Outline

- Neighborhood image processing,
 - Connected Component Analysis (CCA)
 - Mathematical Morphology
 - Dilation
 - Corrosion
 - Opening
 - Closing
- Image filtering
 - Convolution
 - Frequently used filters
 - Gaussian filter
 - Nonlinear Filters
 - Median filter
 - Bilateral filter
- Frequency domain analysis

Mathematical definition of image convolution

- Image $f(x, y)$ and kernel $k(x, y)$
- Continuous form, $(x, y) \in \mathbb{R}^2$

$$k(x, y) * f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} k(u, v) f(x - u, y - v) du dv$$

- Discrete form, kernel size (W_k, H_k) , Image size (W_f, H_f) ,

$$k(x, y) * f(x, y) = \sum_{i=0}^{W_k-1} \sum_{j=0}^{H_k-1} k(i, j) f(x - i, y - j)$$

- Commutative:

$$k(x, y) * f(x, y) = f(x, y) * k(x, y)$$

Mathematical definition of cross correlation

- Image $f(x, y)$ and kernel $k(x, y)$
- Continuous form, $(x, y) \in \mathbb{R}^2$

$$k(x, y) \otimes f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} k(u, v) f(x + u, y + v) du dv$$

- Discrete form, kernel size (W_k, H_k) , Image size (W_f, H_f) ,

$$k(x, y) \otimes f(x, y) = \sum_{i=0}^{W_k-1} \sum_{j=0}^{H_k-1} k(i, j) f(x + i, y + j)$$

- If $k = f$, the operation is called auto-correlation
- Relation with convolution

$$k(x, y) \otimes f(x, y) = k(-x, -y) * f(x, y)$$

Outline

- Neighborhood image processing,
 - Connected Component Analysis (CCA)
 - Mathematical Morphology
 - Dilation
 - Corrosion
 - Opening
 - Closing
- Image filtering
 - Convolution
 - Frequently used filters
 - Gaussian filter
 - Nonlinear Filters
 - Median filter
 - Bilateral filter
- Frequency domain analysis

Image filtering

- Very useful
 - Enhance images
 - De-noise, resize, increase contrast, etc.
 - Extract information from images
 - Texture, edges, distinctive points, etc.
 - Detect patterns
 - Template matching

A taxonomy of useful filters

- Impulse, Shifts,
- Blur
 - Box filter
 - Gaussian filter
 - Bilateral filter
 - Asymmetrical filter: motion blur
- Edges
 - $[-1 \ 1]$
 - Derivative filter
 - Derivative of a gaussian
 - Oriented filters
 - Gabor filter
 - Quadrature filters: phase and magnitude.
 - Elongated edges: filling gaps...

Box Filter

$g[\cdot, \cdot]$

$$\frac{1}{9}$$

1	1	1
1	1	1
1	1	1

Practice with linear filters



Original

0	0	0
0	1	0
0	0	0

?

Practice with linear filters



Original

0	0	0
0	1	0
0	0	0



Filtered
(no change)

Practice with linear filters



Original

0	0	0
0	0	1
0	0	0

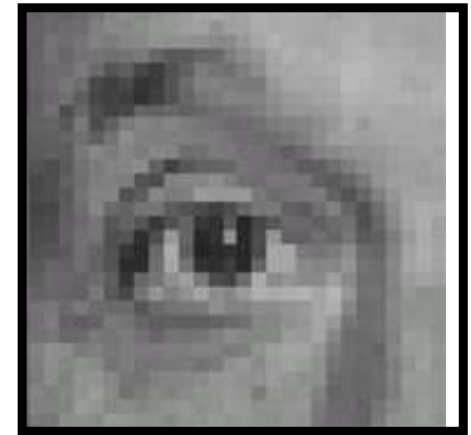
?

Practice with linear filters



Original

0	0	0
0	0	1
0	0	0



Shifted left
By 1 pixel

Practice with linear filters



Original

0	0	0
0	2	0
0	0	0

—

$\frac{1}{9}$

1	1	1
1	1	1
1	1	1

?

(Note that filter sums to 1)

Practice with linear filters



Original

0	0	0
0	2	0
0	0	0

−

$\frac{1}{9}$

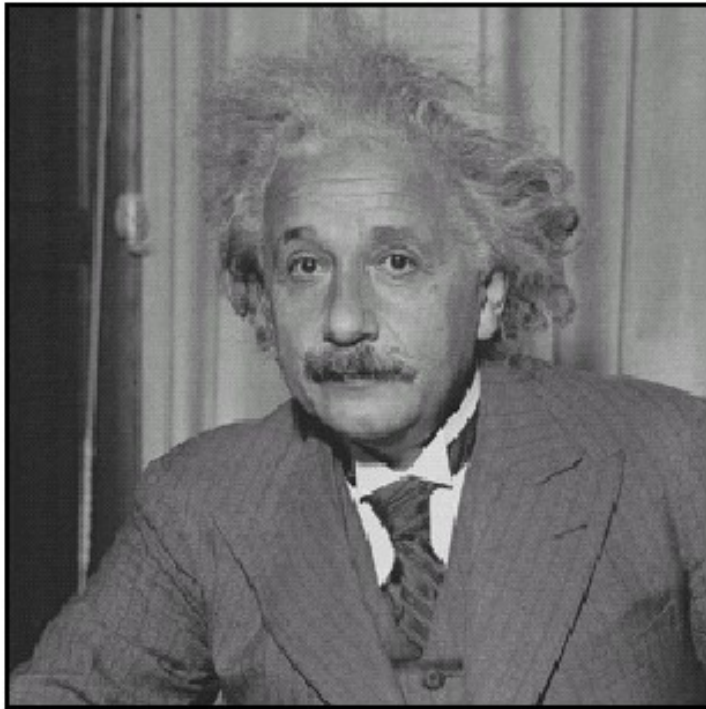
1	1	1
1	1	1
1	1	1



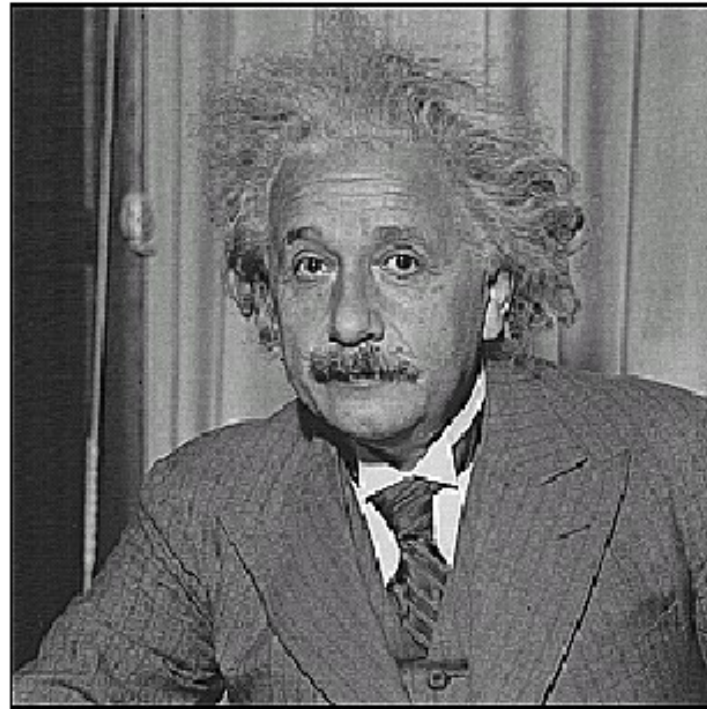
Sharpening filter

- Accentuates differences with local average

Sharpening



before



after

Edge filter: $[-1 \ 1]$

$$\frac{\partial I}{\partial x} \simeq I(x, y) - I(x - 1, y)$$



$g[m,n]$

$\otimes$

$[-1, 1]$

$=$

$h[m,n]$



$f[m,n]$

$$[-1 \ 1]^T$$



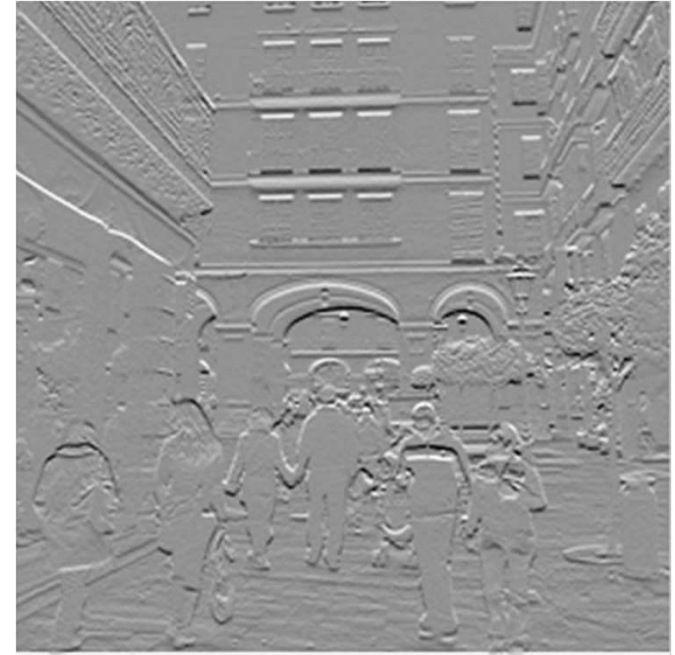
$g[m,n]$



$$[-1, 1]^T$$

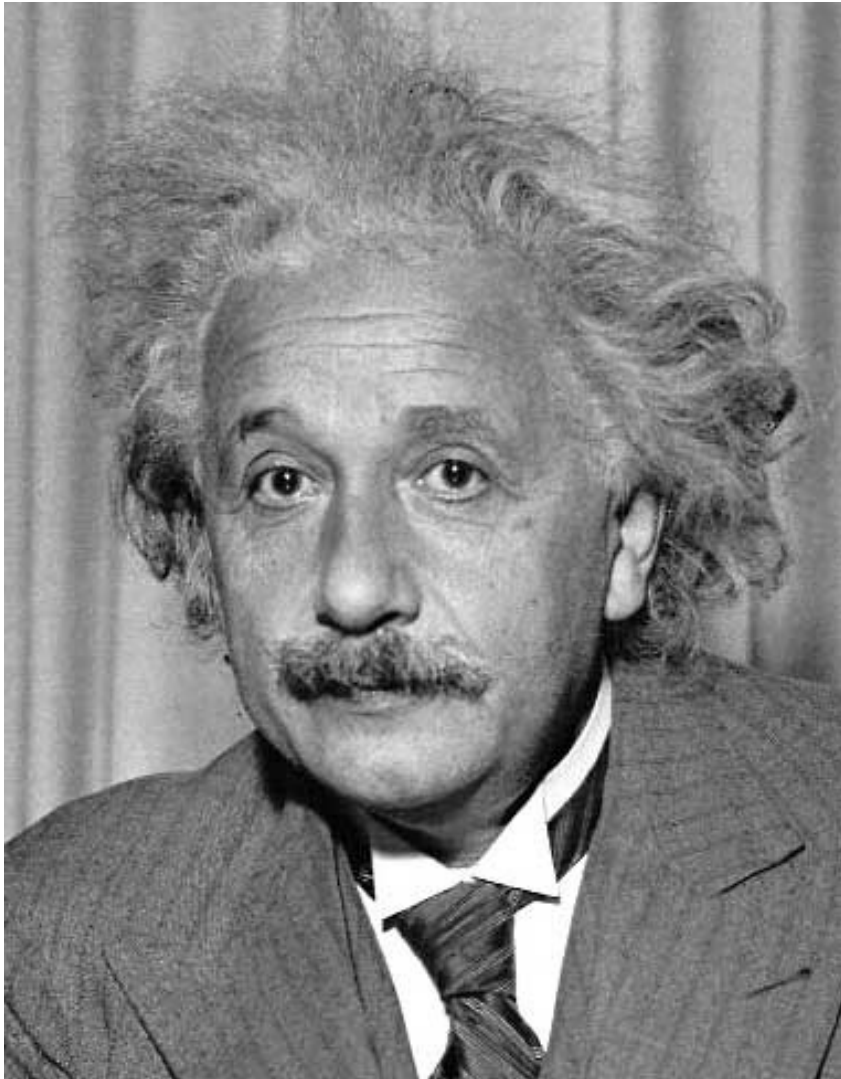
=

$$h[m,n]$$



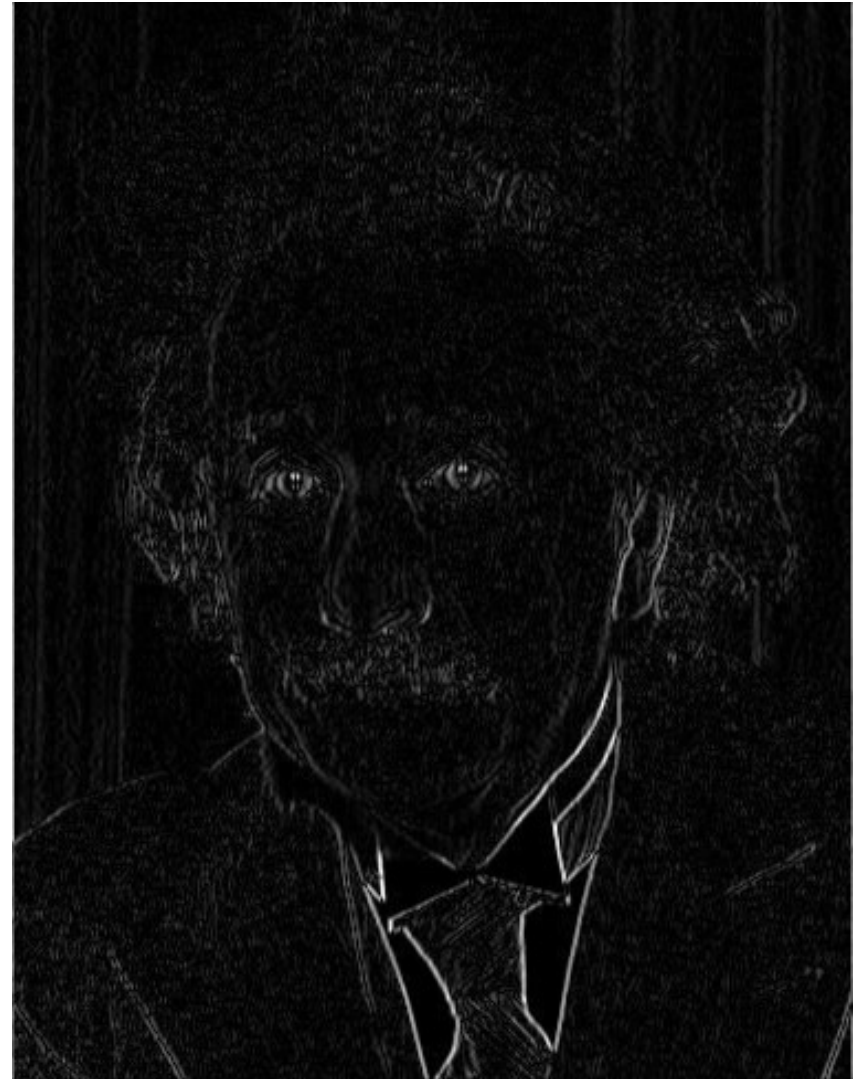
$f[m,n]$

Derivative filters



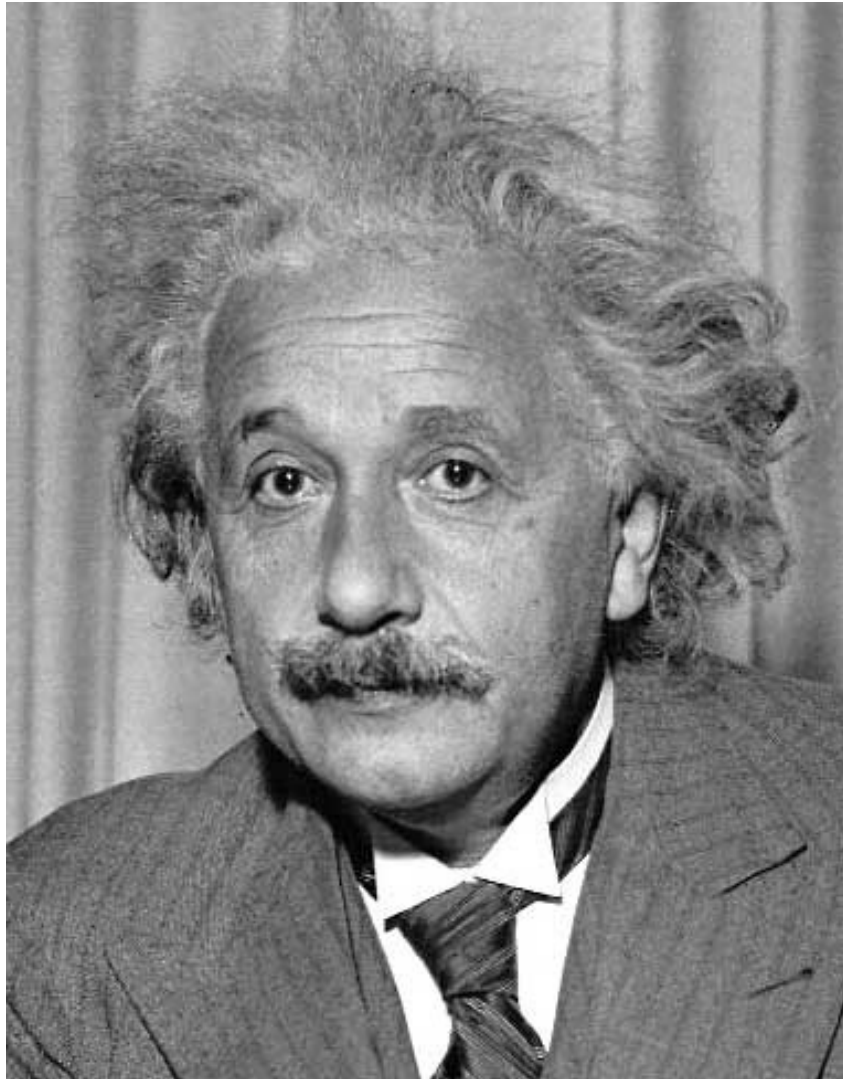
1	0	-1
2	0	-2
1	0	-1

Sobel



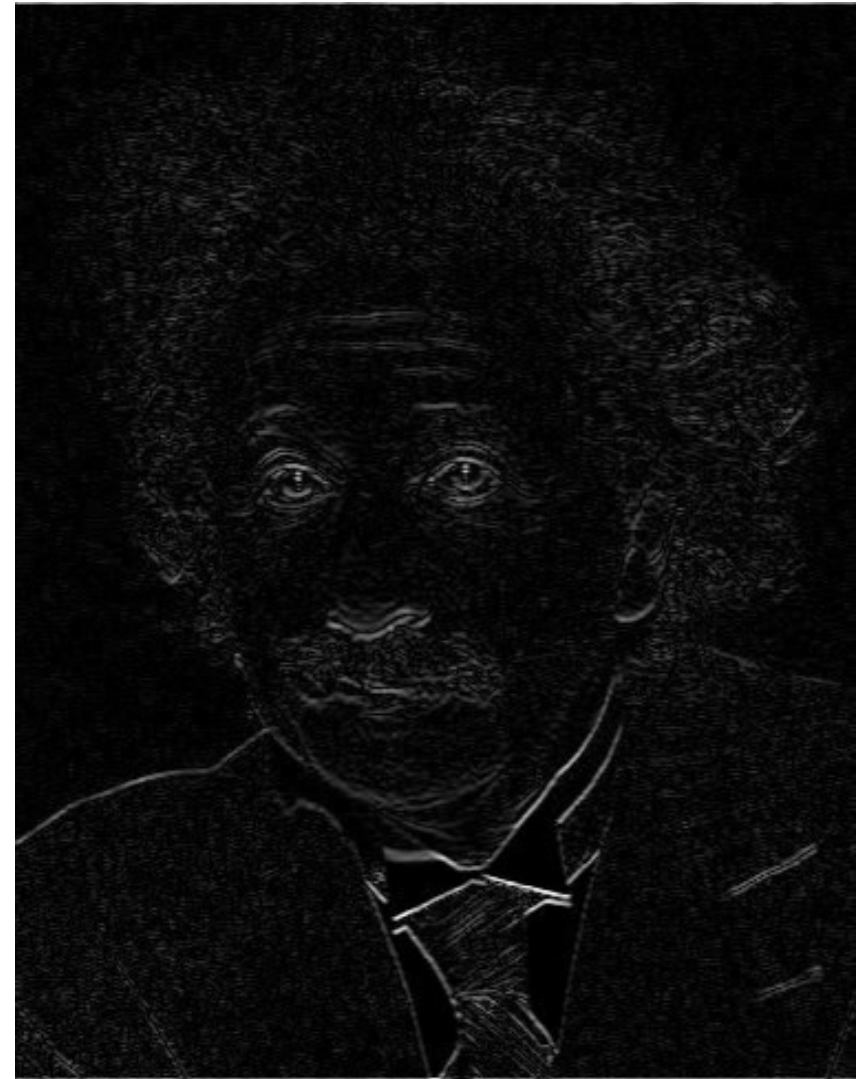
Vertical Edge
(absolute value)

Derivative filters



1	2	1
0	0	0
-1	-2	-1

Sobel



Horizontal Edge
(absolute value)

Motion blur filter



$g[m,n]$



$h[m,n]$



$f[m,n]$

Motion blur filter



$g[m,n]$



$h[m,n]$



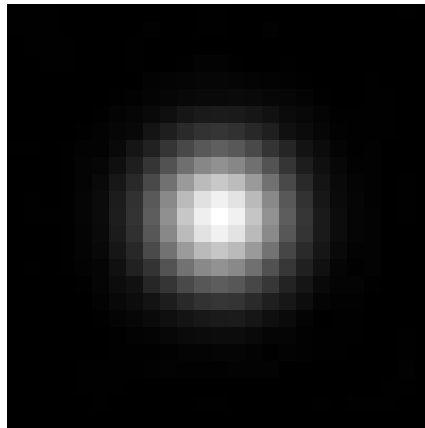
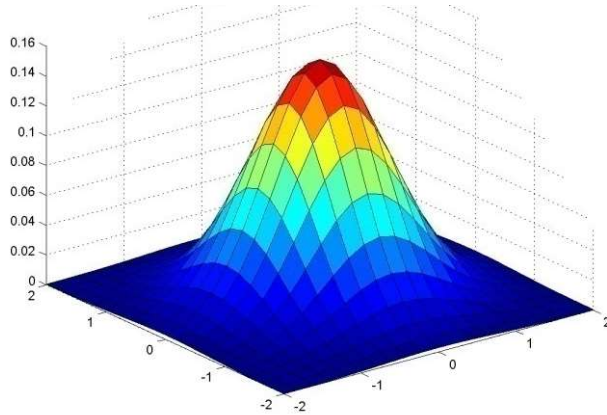
$f[m,n]$

Outline

- Neighborhood image processing,
 - Connected Component Analysis (CCA)
 - Mathematical Morphology
 - Dilation
 - Corrosion
 - Opening
 - Closing
- Image filtering
 - Convolution
 - Frequently used filters
 - Gaussian filter
 - Nonlinear Filters
 - Median filter
 - Bilateral filter
- Frequency domain analysis

Important linear filter: Gaussian

- Weight contributions of neighboring pixels by nearness



0.003	0.013	0.022	0.013	0.003
0.013	0.059	0.097	0.059	0.013
0.022	0.097	0.159	0.097	0.022
0.013	0.059	0.097	0.059	0.013
0.003	0.013	0.022	0.013	0.003

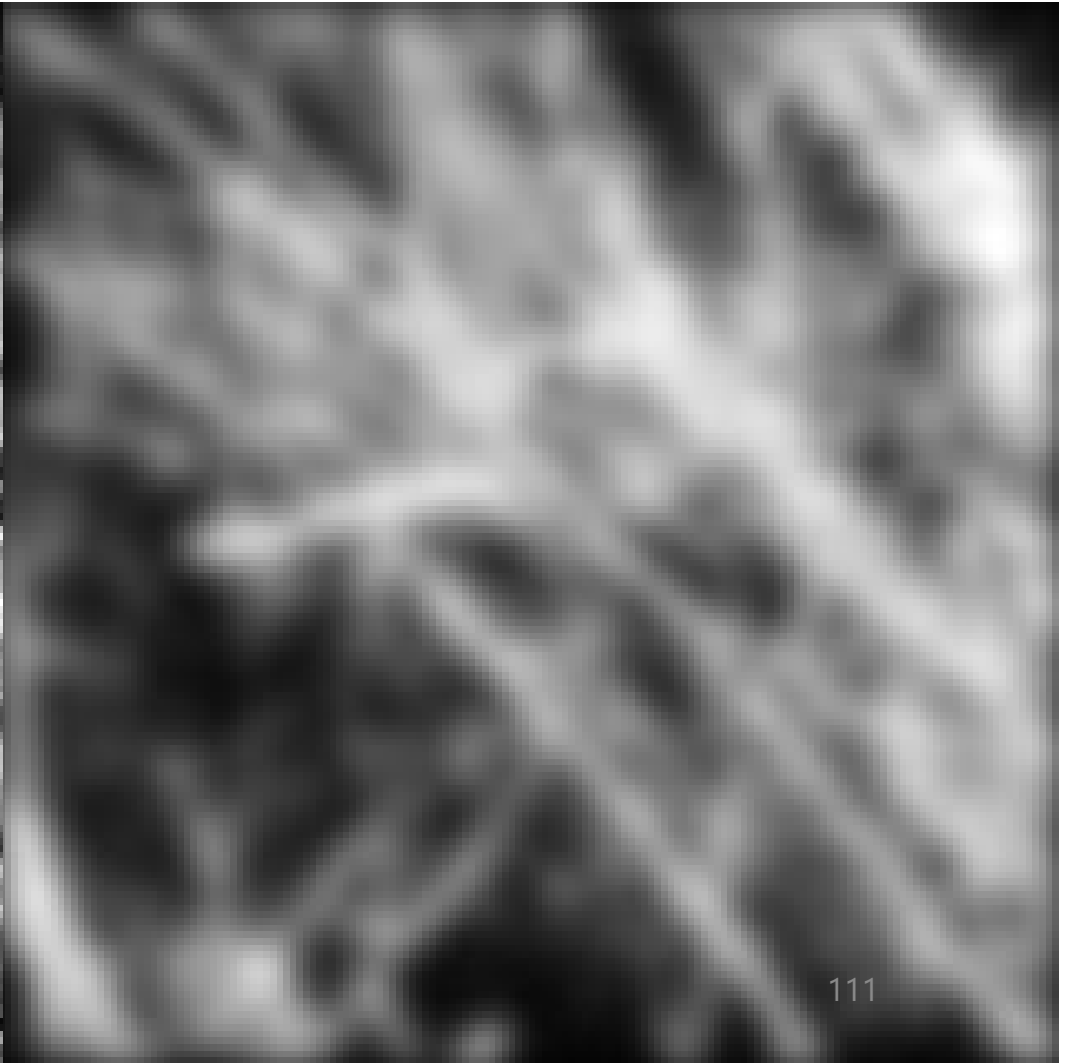
5 x 5, $\sigma = 1$

$$G_{\sigma} = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$

Smoothing with box filter

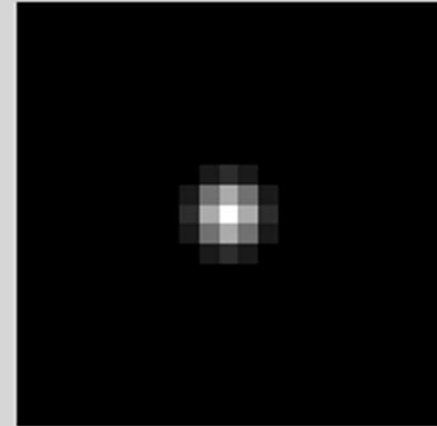
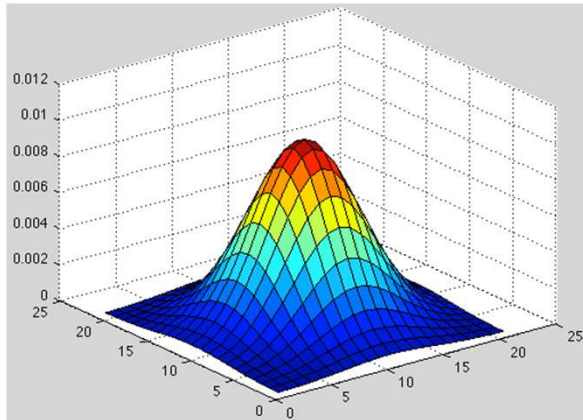


Smoothing with Gaussian filter

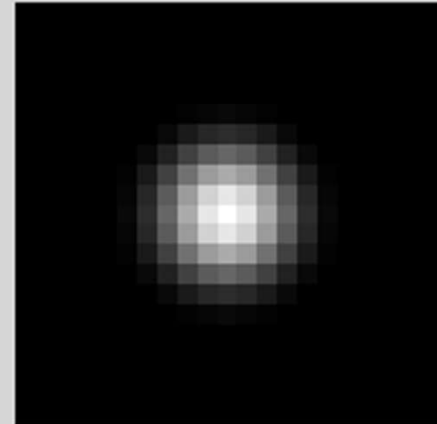


Gaussian filter: change scales

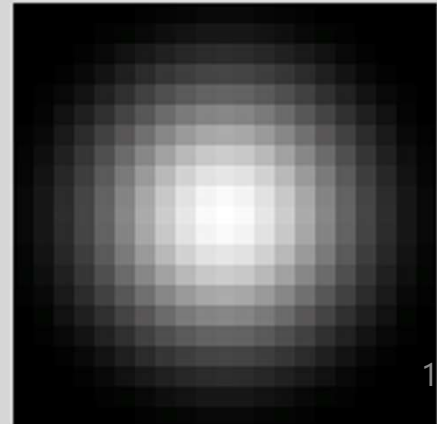
$$G(x, y; \sigma) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$



$\sigma=1$



$\sigma=2$



$\sigma=4$

Properties of Gaussian filter

- Rotational symmetry treats features of all orientations equally (isotropy).
- Efficient: Rule of thumb is kernel width $\geq 5\sigma$
 - Separable
 - Cascadable: Approach to large σ comes from identity

$$G_{\sigma_1} * G_{\sigma_2} = G_{\sqrt{\sigma_1^2 + \sigma_2^2}}$$

Separability of the Gaussian filter

$$\begin{aligned} G_{\sigma}(x, y) &= \frac{1}{2\pi\sigma^2} \exp^{-\frac{x^2 + y^2}{2\sigma^2}} \\ &= \left(\frac{1}{\sqrt{2\pi}\sigma} \exp^{-\frac{x^2}{2\sigma^2}} \right) \left(\frac{1}{\sqrt{2\pi}\sigma} \exp^{-\frac{y^2}{2\sigma^2}} \right) \end{aligned}$$

The 2D Gaussian can be expressed as the product of two functions, one a function of x and the other a function of y

In this case, the two functions are the (identical) 1D Gaussian

Separability example

2D convolution
(center location only)

$$\begin{array}{|c|c|c|} \hline 1 & 2 & 1 \\ \hline 2 & 4 & 2 \\ \hline 1 & 2 & 1 \\ \hline \end{array} * \begin{array}{|c|c|c|} \hline 2 & 3 & 3 \\ \hline 3 & 5 & 5 \\ \hline 4 & 4 & 6 \\ \hline \end{array}$$

The filter factors
into a product of 1D
filters:

$$\begin{array}{|c|c|c|} \hline 1 & 2 & 1 \\ \hline 2 & 4 & 2 \\ \hline 1 & 2 & 1 \\ \hline \end{array} = \begin{array}{|c|} \hline 1 \\ \hline 2 \\ \hline 1 \\ \hline \end{array} \times \begin{array}{|c|c|c|} \hline 1 & 2 & 1 \\ \hline \end{array}$$

Perform convolution
along rows:

$$\begin{array}{|c|c|c|} \hline 1 & 2 & 1 \\ \hline \end{array} * \begin{array}{|c|c|c|} \hline 2 & 3 & 3 \\ \hline 3 & 5 & 5 \\ \hline 4 & 4 & 6 \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline & 11 & \\ \hline & 18 & \\ \hline & 18 & \\ \hline \end{array}$$

Followed by convolution
along the remaining column:

Outline

- Neighborhood image processing,
 - Connected Component Analysis (CCA)
 - Mathematical Morphology
 - Dilation
 - Corrosion
 - Opening
 - Closing
- Image filtering
 - Convolution
 - Frequently used filters
 - Gaussian filter
 - Nonlinear Filters
 - Median filter
 - Bilateral filter
- Frequency domain analysis

Pepper salt noise

Gaussian

3x3



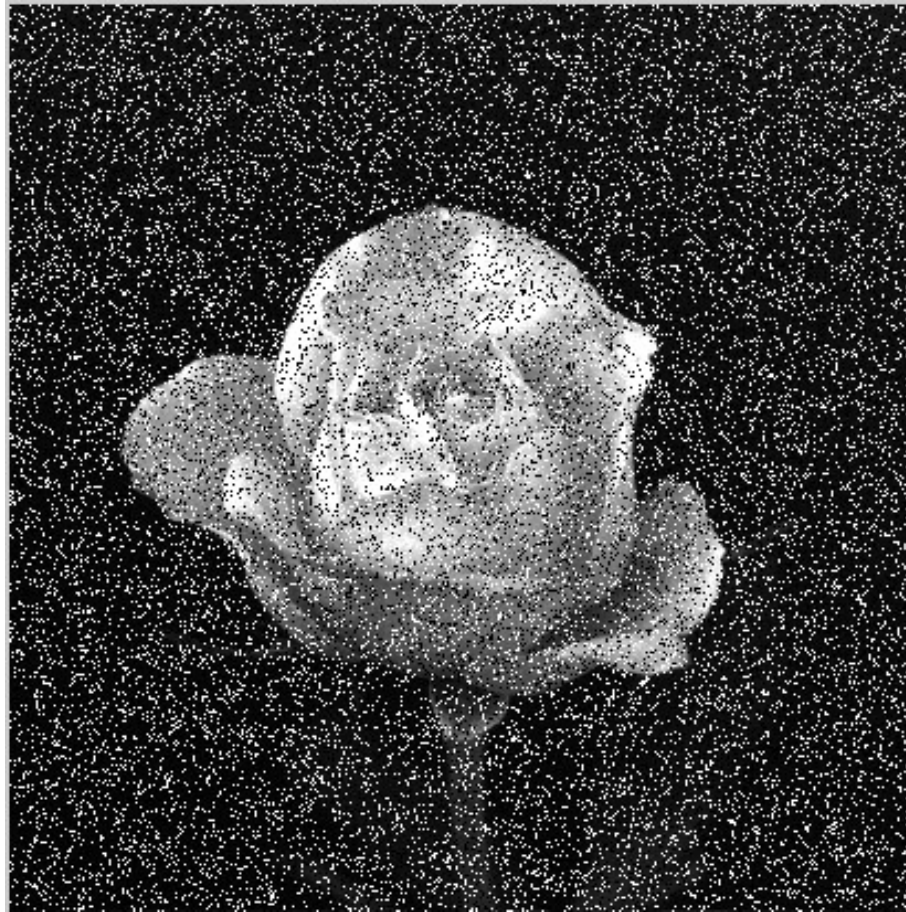
5x5



7x7



How about this?



Median

1, 5, 2, 80, 7.

1, 2, 5, 7, 80

Mean = 19

Median = 5

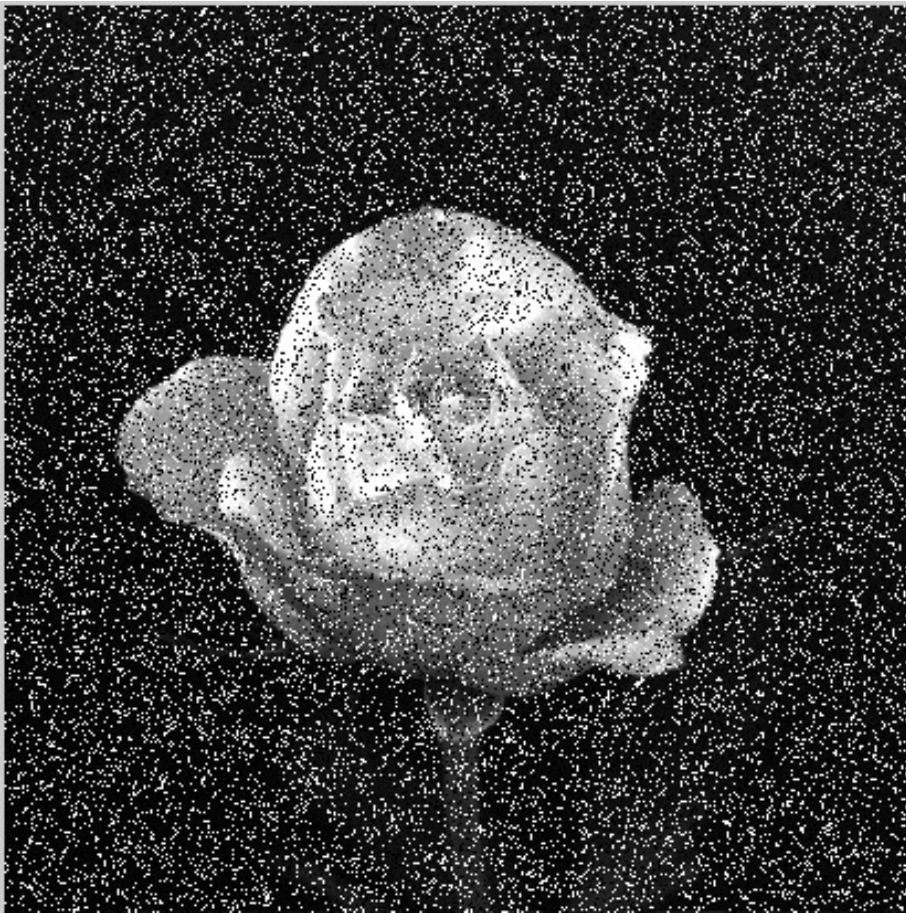
Questions

What is the value of the yellow box after 3x3 median filtering?

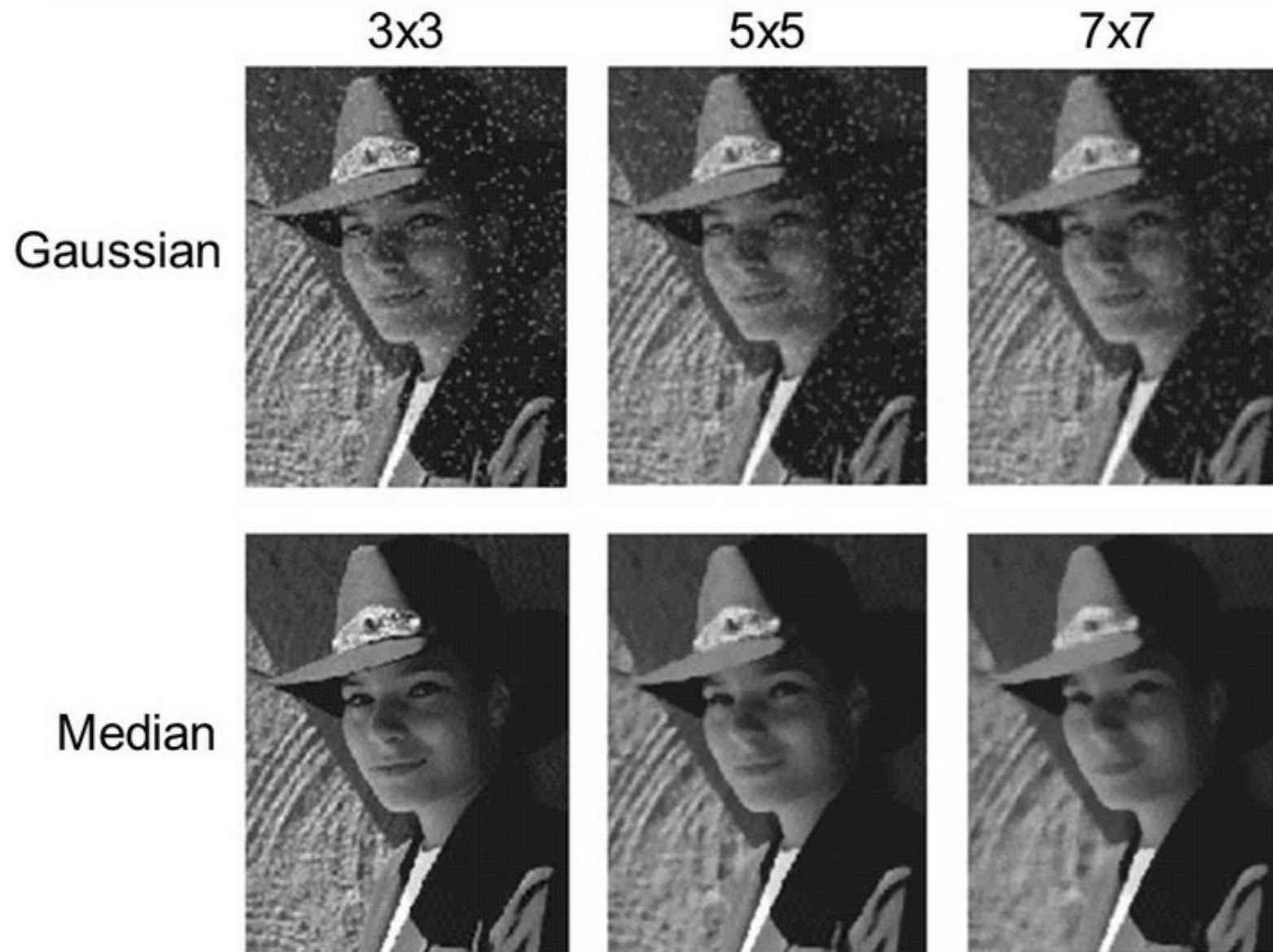
1	2	2	2	3
2	5	2	4	3
2	2	2	2	3
4	3	7	5	3
3	3	3	1	1

2,2,2,
2,3,4,
5,5,7

Example: Median filter result



Gaussian vs. median filtering



Outline

- Neighborhood image processing,
 - Connected Component Analysis (CCA)
 - Mathematical Morphology
 - Dilation
 - Corrosion
 - Opening
 - Closing
- Image filtering
 - Convolution
 - Frequently used filters
 - Gaussian filter
 - Nonlinear Filters
 - Median filter
 - Bilateral filter
- Frequency domain analysis

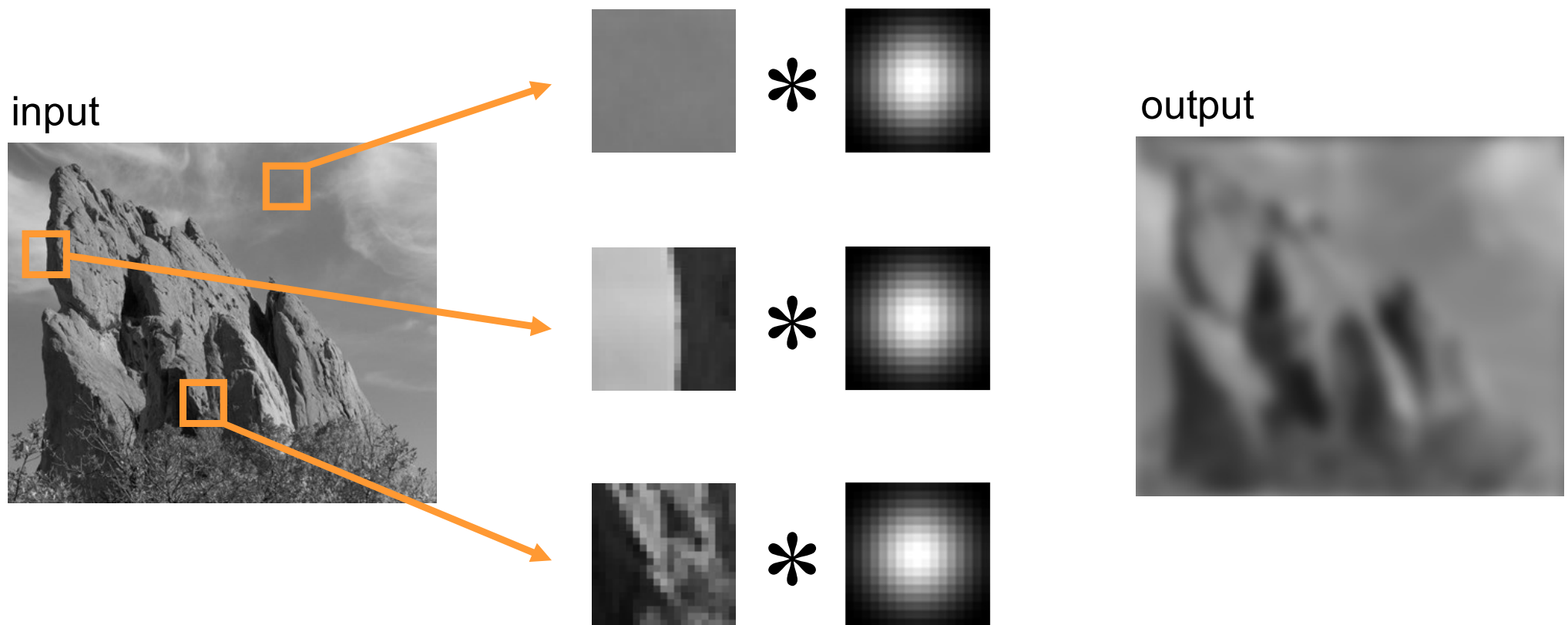
How to smooth an image without causing excessive edge-blur ?

Bilateral Filter (Tomasi et al.1998)

- Smooths regions while preserving edges

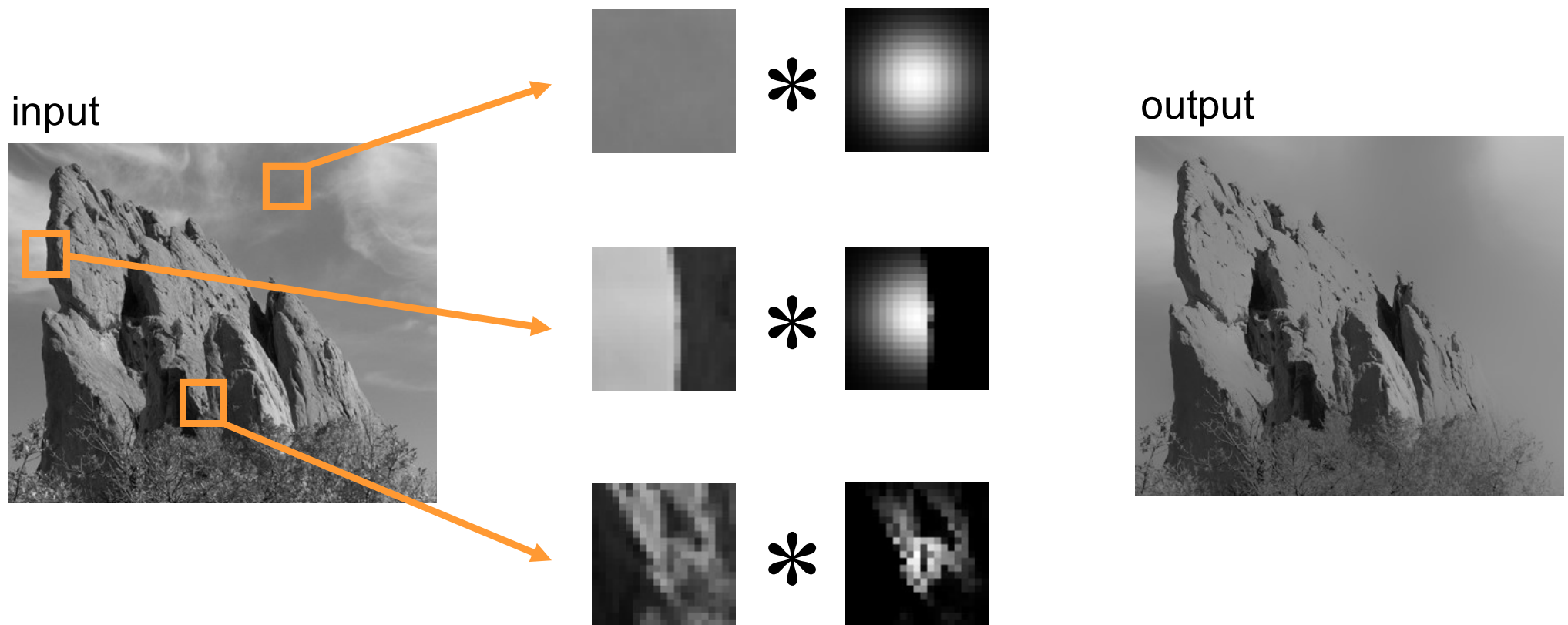


Blur Comes from Averaging across Edges



Bilateral Filter

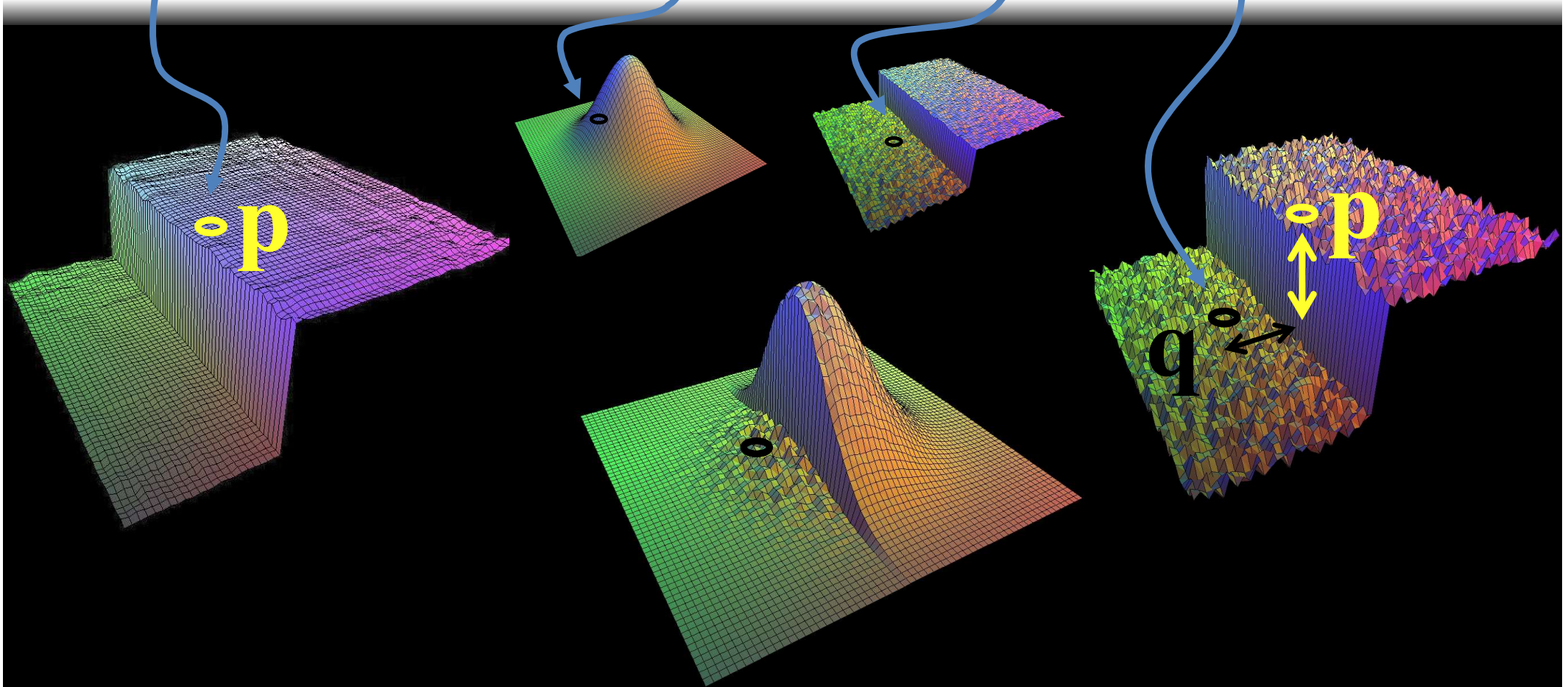
No Averaging across Edges



The kernel shape depends on the image content.

Bilateral Filter on a Height Field

$$\underbrace{BF[I]_p}_{\text{Input}} = \frac{1}{W_p} \sum_{q \in S} \underbrace{G_{\sigma_s}(\|\mathbf{p} - \mathbf{q}\|)}_{\text{Spatial}} \underbrace{G_{\sigma_r}(|I_p - I_q|)}_{\text{Range}} \underbrace{I_q}_{\text{Output}}$$



Varying the Space Parameter



input

$$\sigma_r = 0.1$$



$$\sigma_s = 2$$

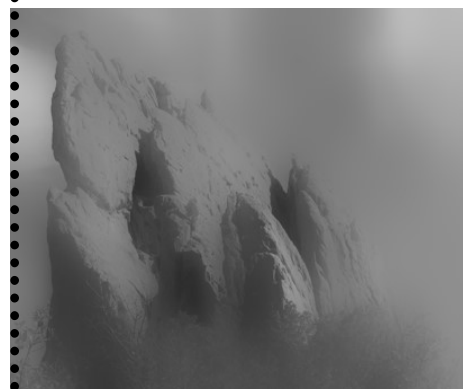
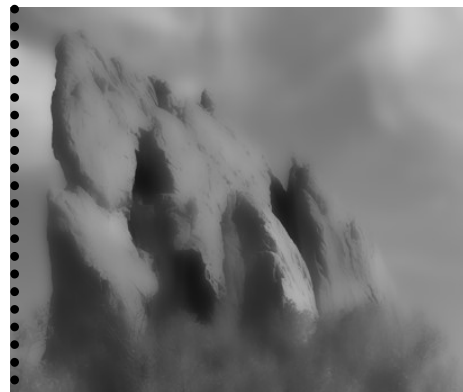


$$\sigma_s = 6$$



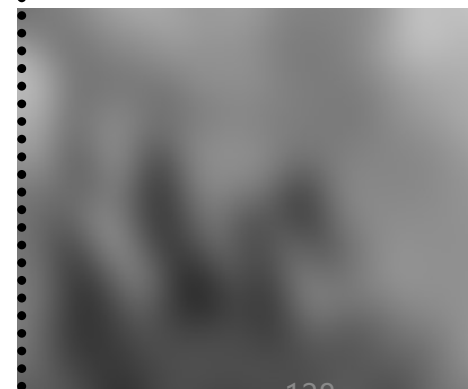
$$\sigma_s = 18$$

$$\sigma_r = 0.25$$



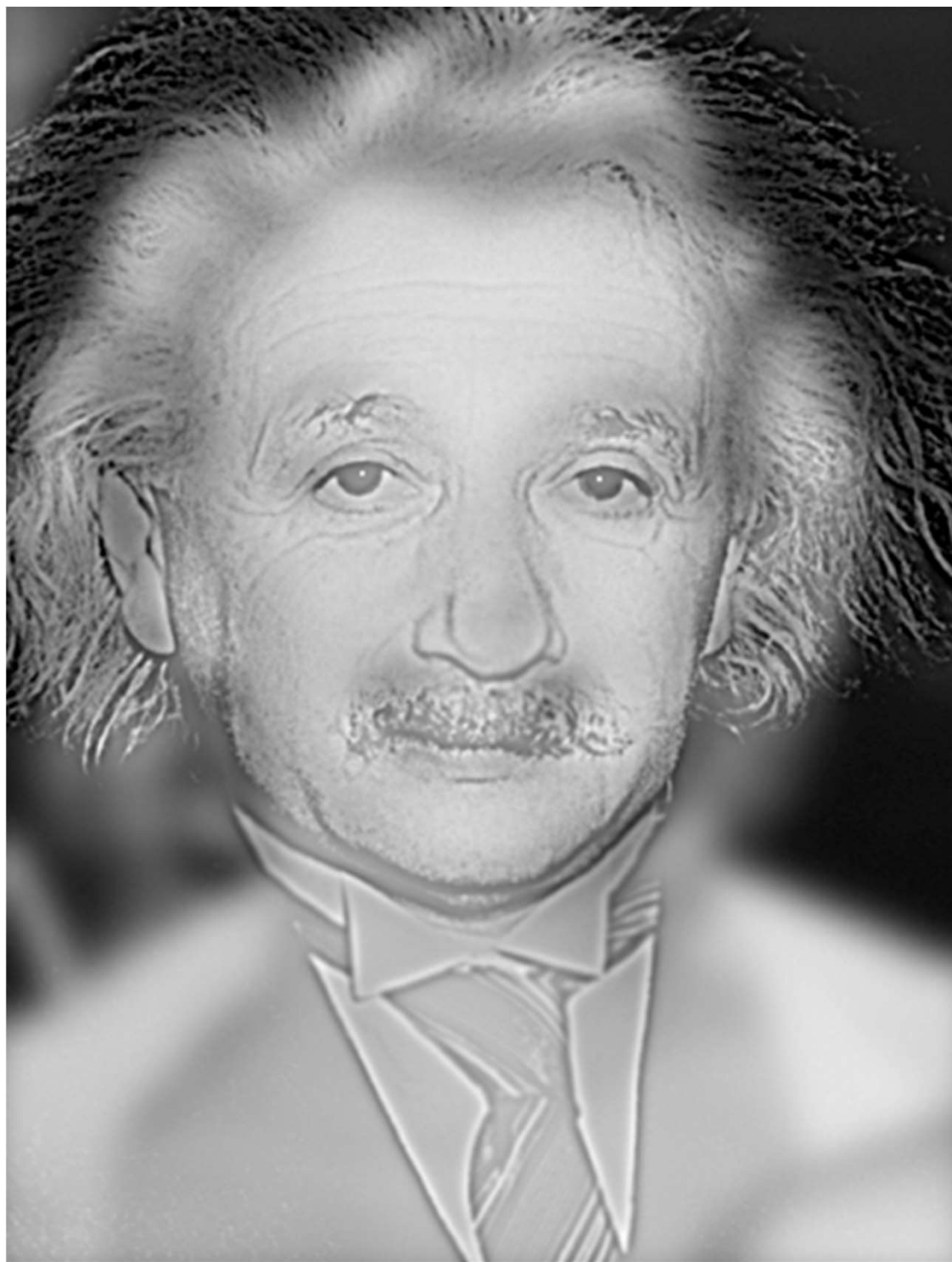
$$\sigma_r = \infty$$

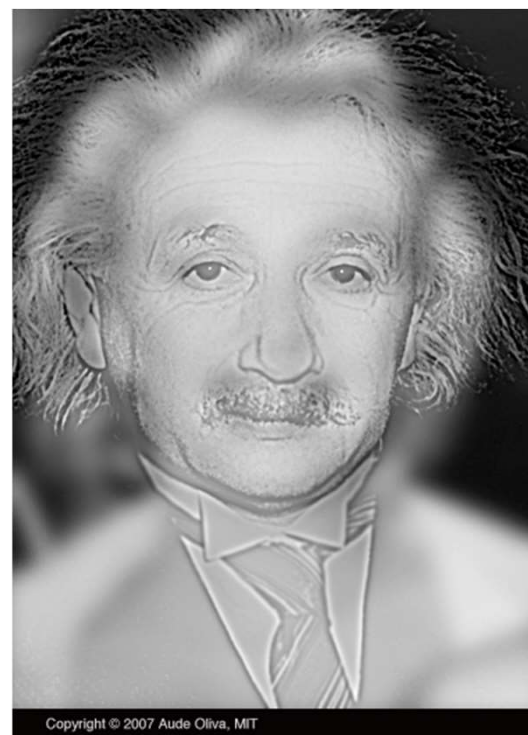
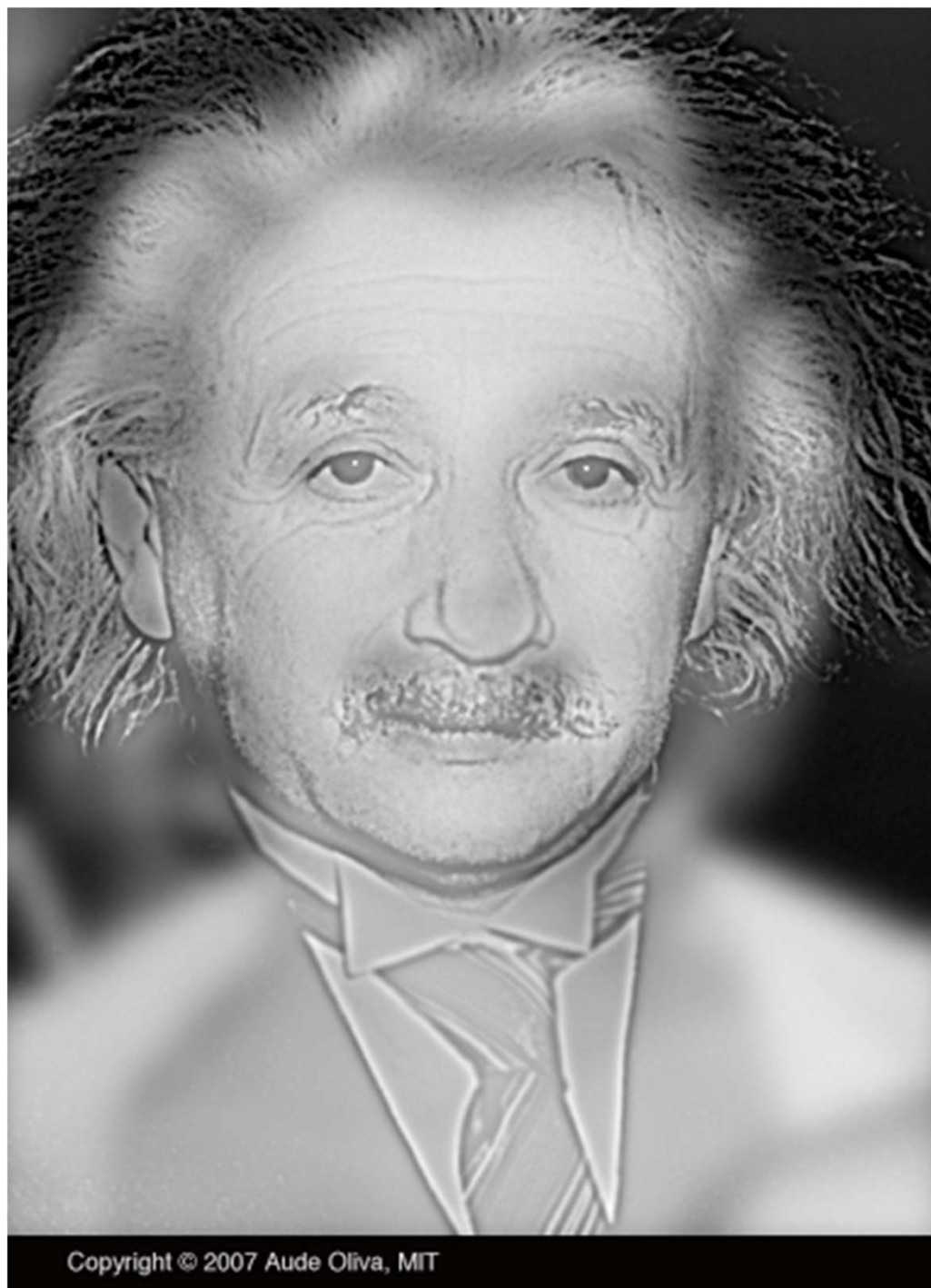
(Gaussian blur)



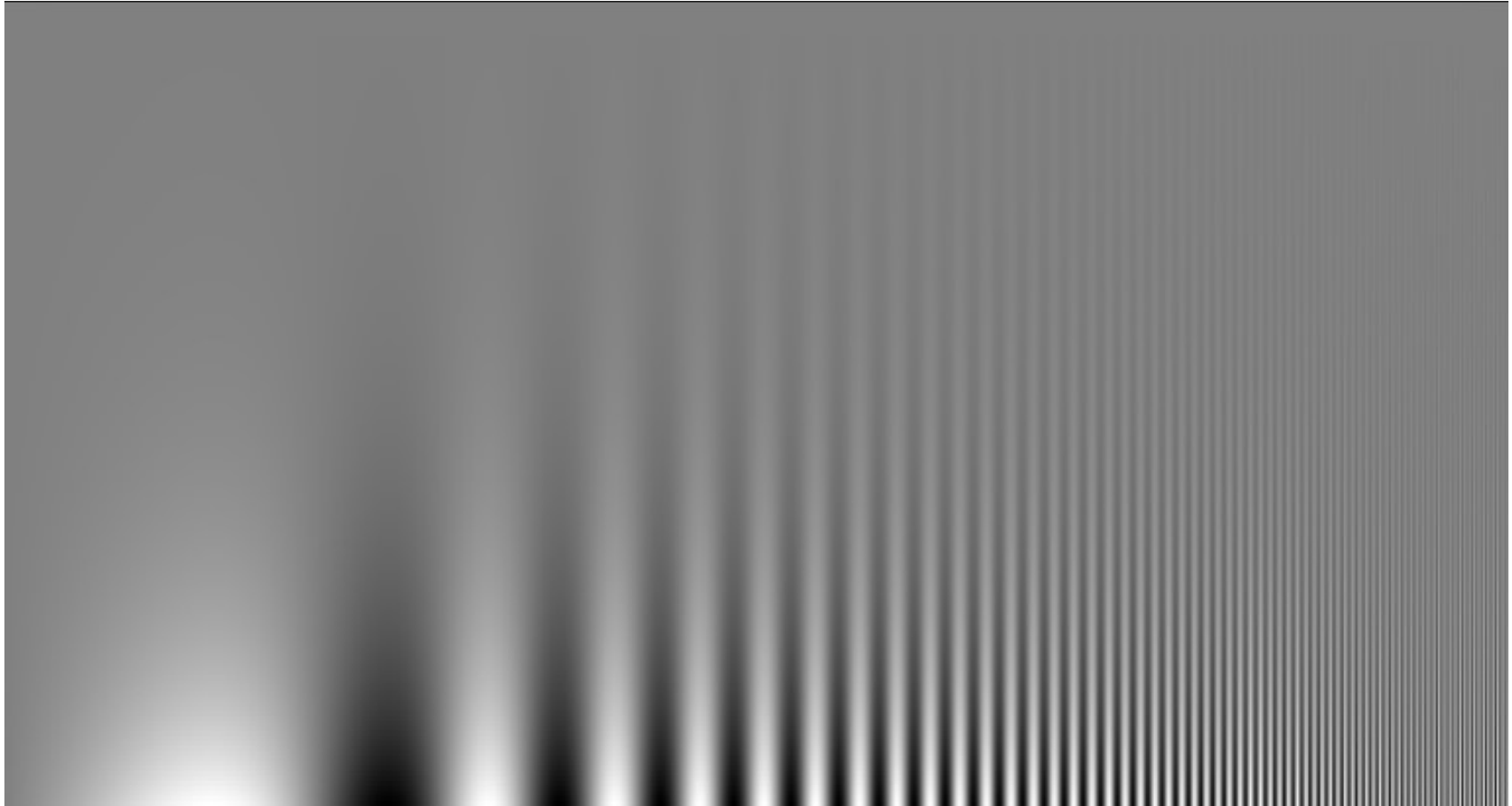
Outline

- Neighborhood image processing
 - Connected Component Analysis (CCA)
 - Mathematical Morphology
 - Dilation
 - Erosion
 - Opening
 - Closing
- Image filtering
 - Convolution
 - Frequently used filters
 - Gaussian filter
 - Nonlinear Filters
 - Median filter
 - Bilateral filter
- Frequency domain analysis

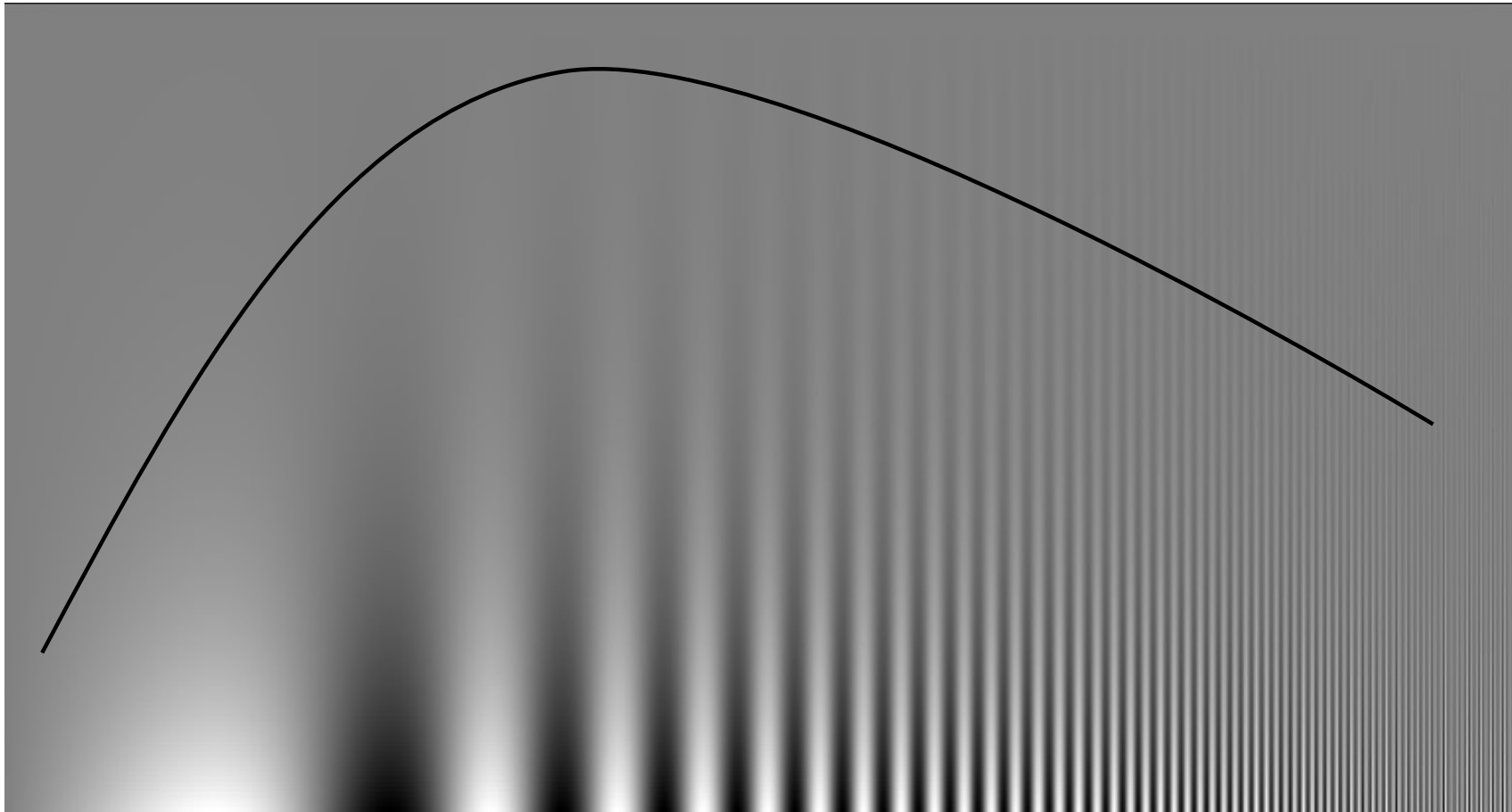




Robson-Campbell Curve



Robson-Campbell Curve



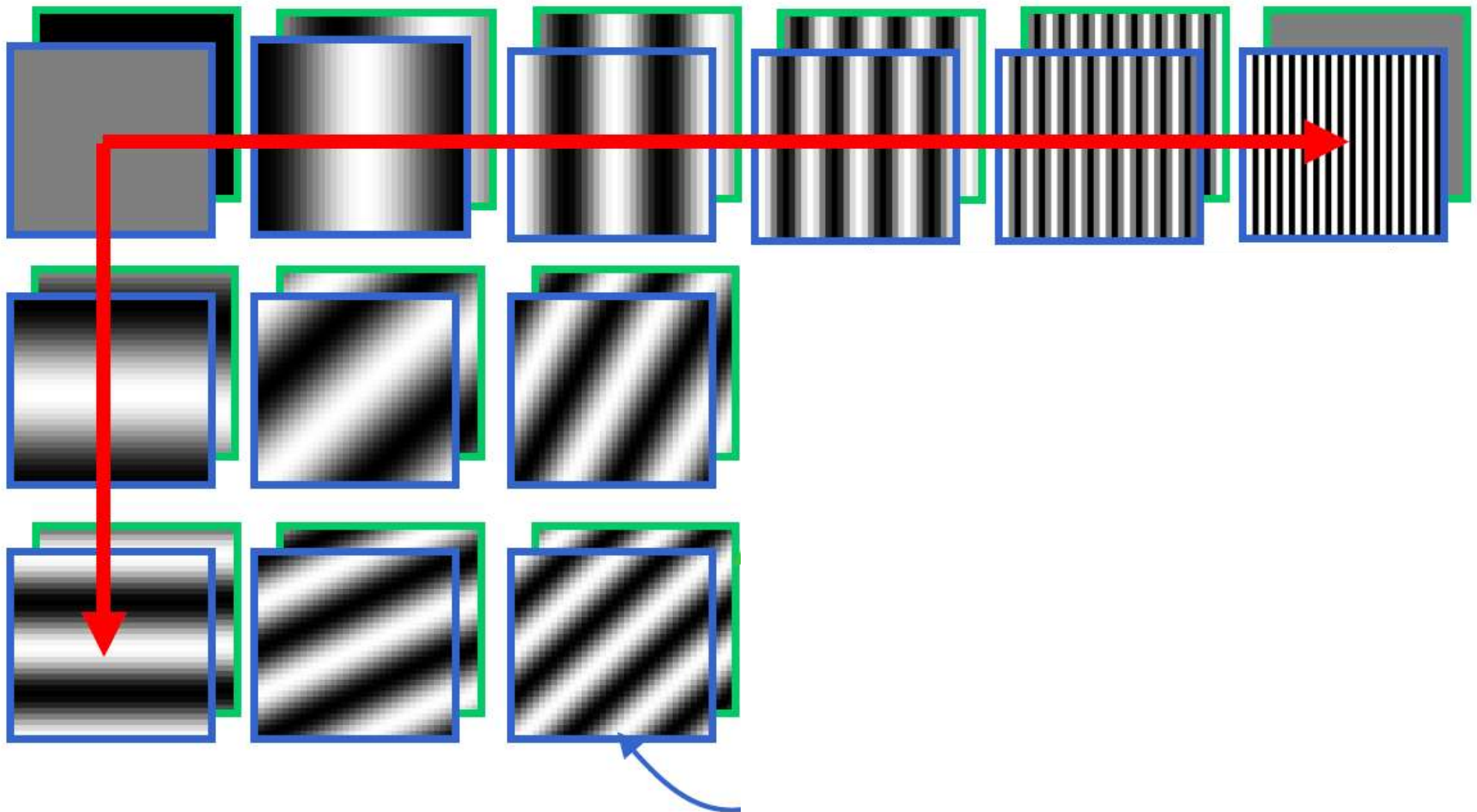
The Discrete Fourier Transform

The *Discrete Fourier Transform* of $f(x, y)$, for $x = 0, 1, 2 \dots M-1$ and $y = 0, 1, 2 \dots N-1$, denoted by $F(u, v)$, is given by the equation:

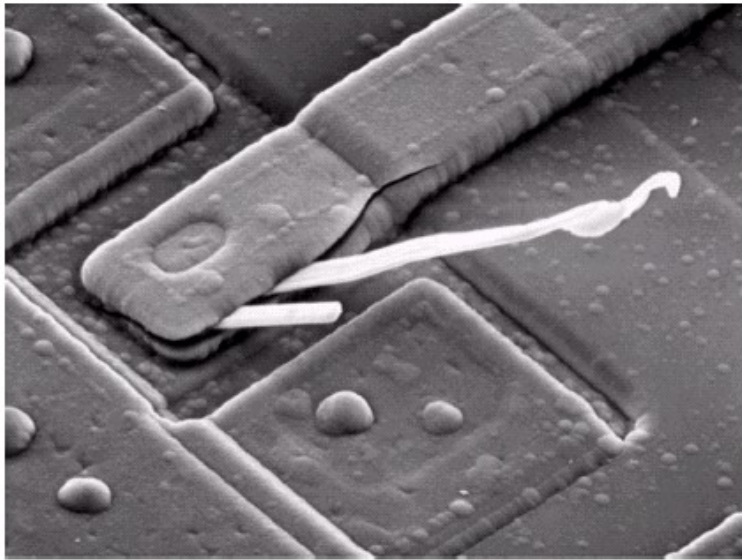
$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

for $u = 0, 1, 2 \dots M-1$ and $v = 0, 1, 2 \dots N-1$.

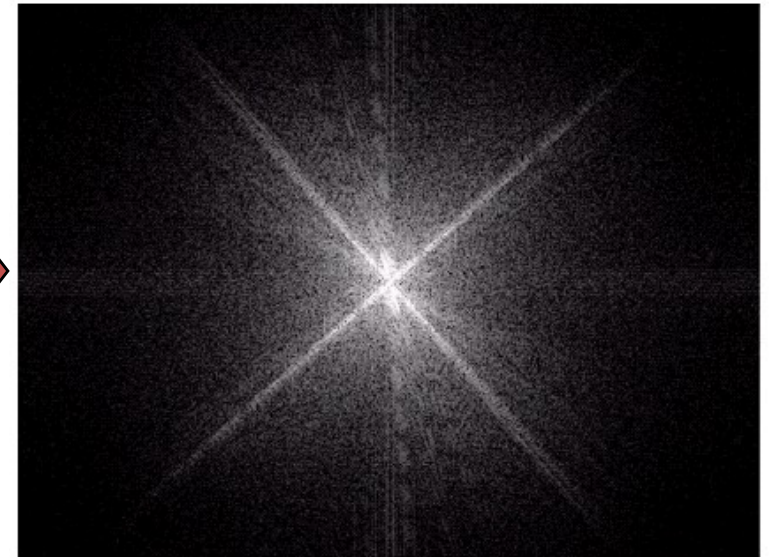
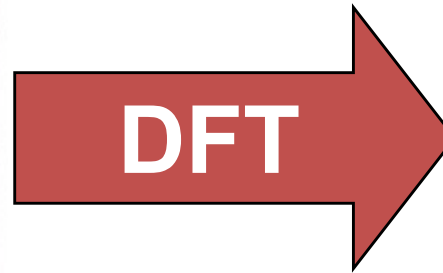
Represent an image by a nice set of Fourier basis



DFT & Images (cont...)

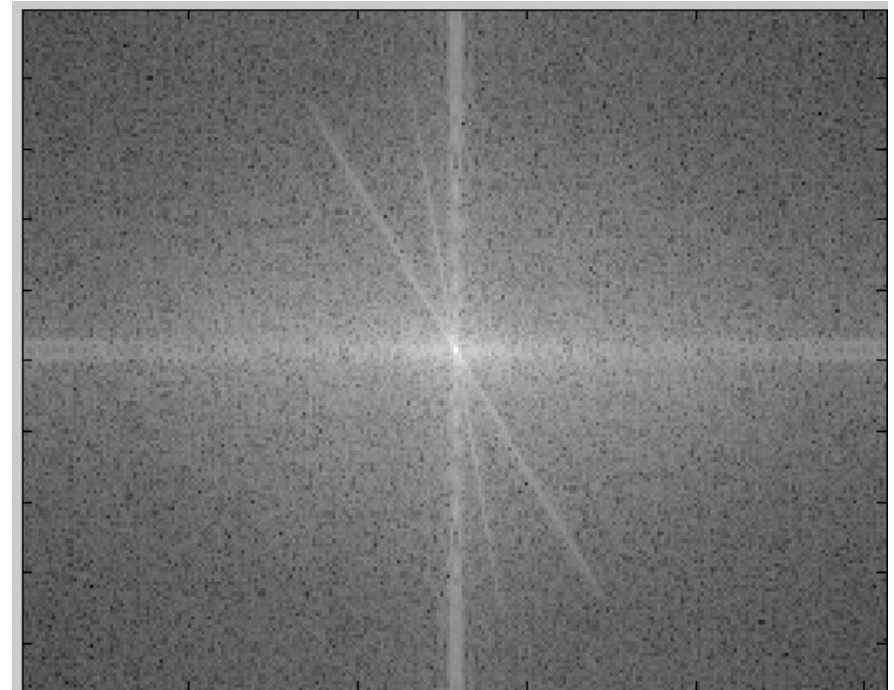


Scanning electron microscope image of an integrated circuit magnified ~2500 times



Fourier spectrum of the image

Man-made Scene



Low pass Filter

spatial domain

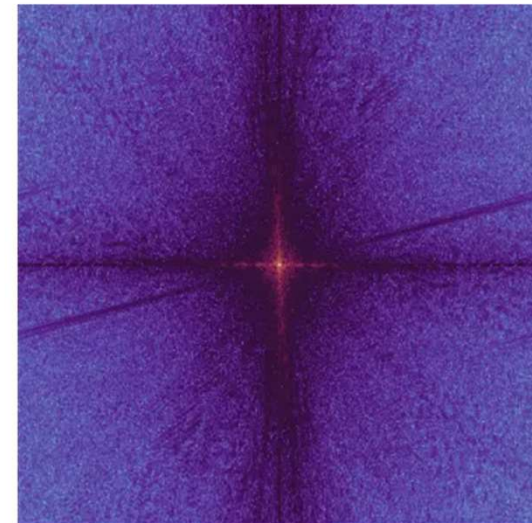
frequency domain

$$f(x,y) \longrightarrow F(u,v)$$

$$G(u,v) = F(u,v) \cdot H(u,v)$$

filter

$$g(x,y) \longleftarrow G(u,v)$$

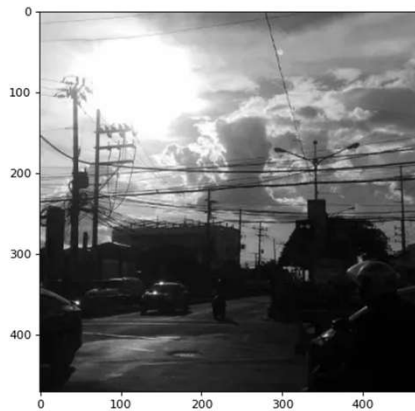


Low pass Filter

Example from Tonichi Edeza

spatial domain

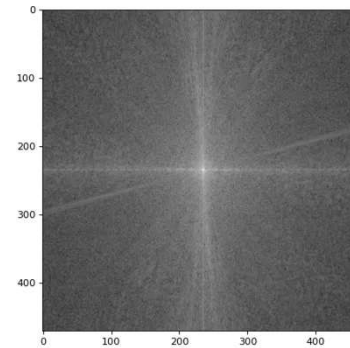
$f(x,y)$



Dark horizontal lines

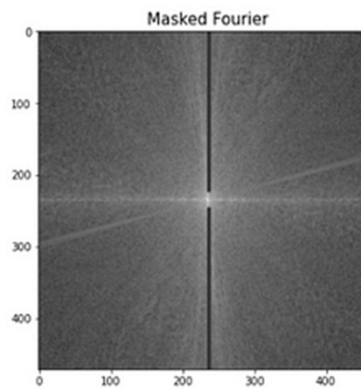
Frequency domain

$F(u,v)$

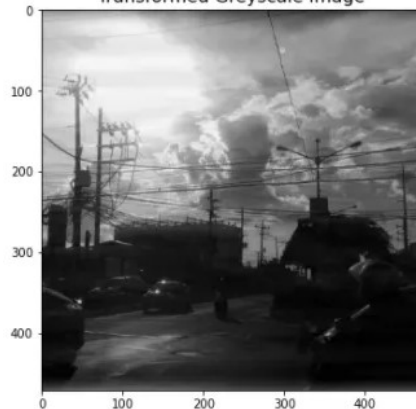


$$G(u,v) = F(u,v) \cdot H(u,v)$$

filter



Transformed Greyscale Image



Thinner horizontal lines

$g(x,y)$

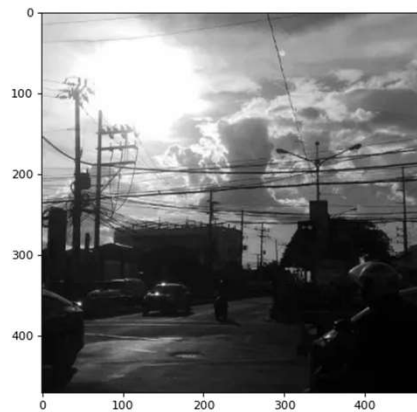
$G(u,v)$

Low pass Filter

Example from Tonichi Edeza

spatial domain

$f(x,y)$



Dark vertical poles

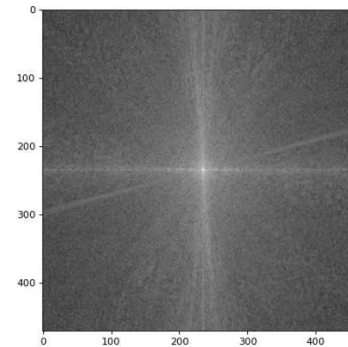


Lighter vertical poles

$g(x,y)$

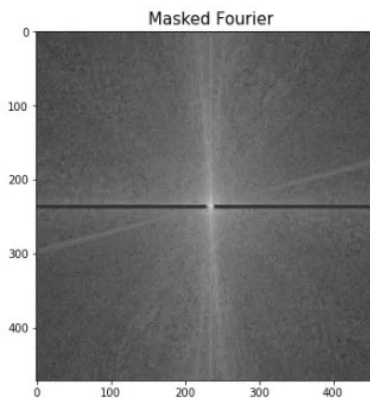
Frequency domain

$F(u,v)$



$$G(u,v) = F(u,v) \cdot H(u,v)$$

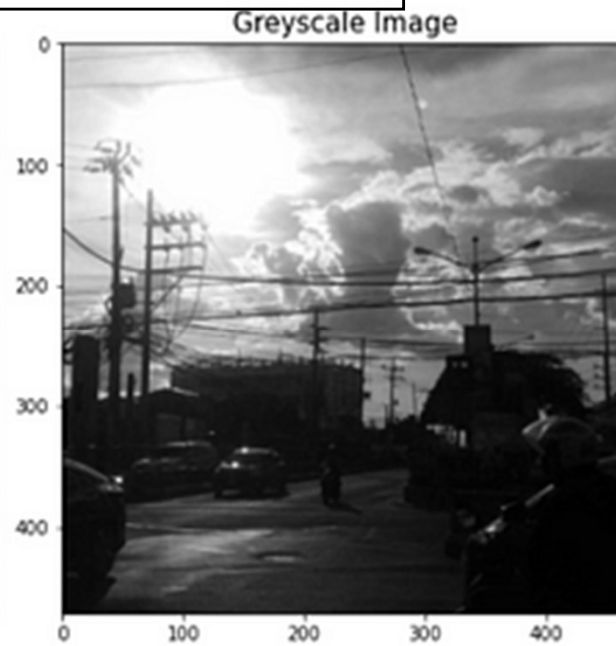
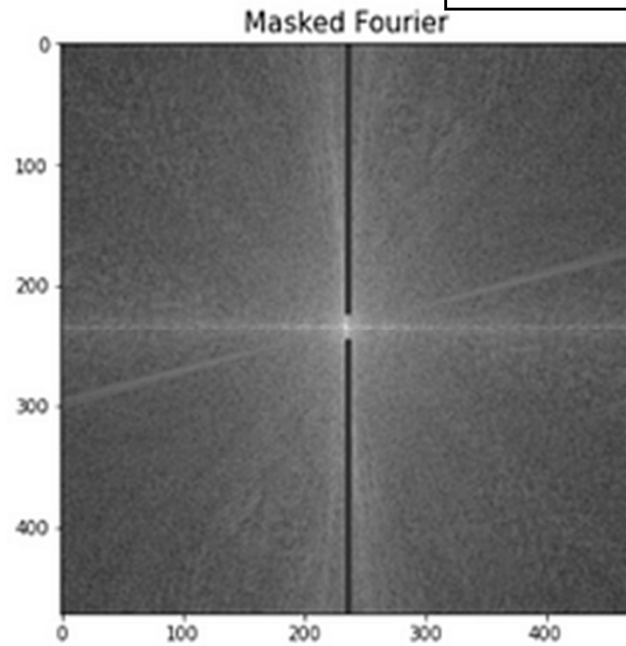
filter



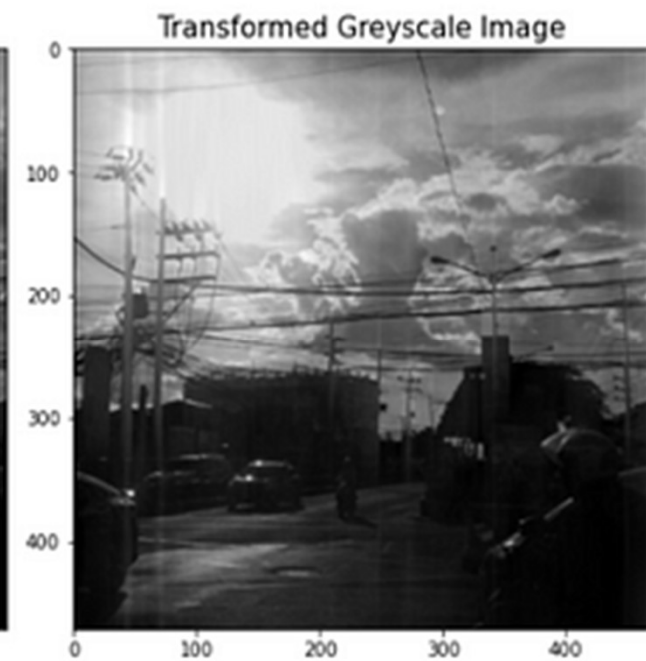
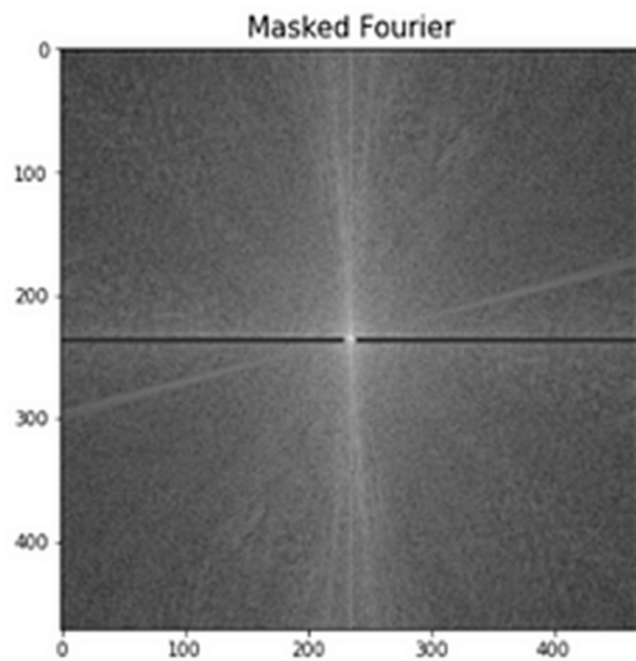
$G(u,v)$

Low pass Filter

Example from Tonichi Edeza

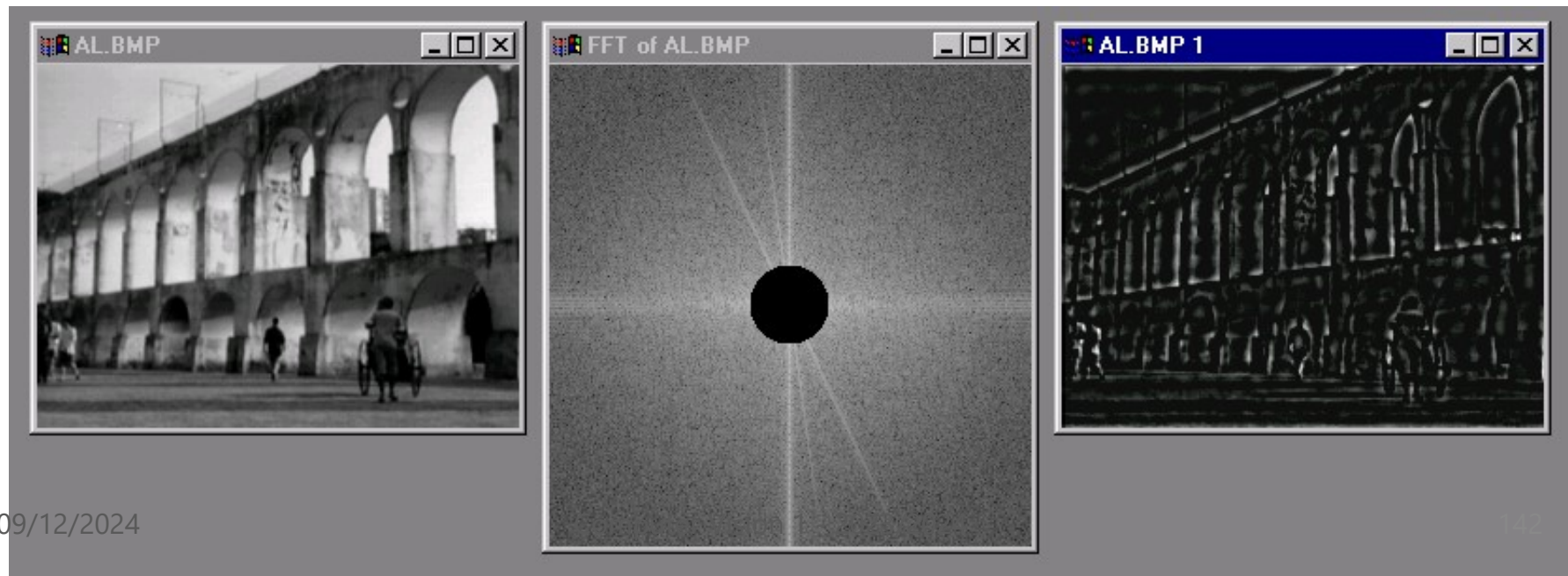


Thinner horizontal lines



Lighter vertical poles

Low and High Pass filtering



Action Points

- Finish Assignment 1, attend labs for help or ask on Canvas Ed Discussion (don't be too late 😊)
- Assignment 2 will be release tomorrow.
- Read textbook and slides for lecture 4 and 5,
- Prepare for exercise week 3.

Disclaimer

- Many of the slides used here are obtained from online resources (including many open lecture materials) without appropriate acknowledgement. They are used here for the sole purpose of classroom teaching. All the credit and all the copyrights belong to the original authors. You should not copy it, redistribute it, put it online, or use it for any other purposes than for this course.